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Middle Permian paleomagnetism of the Sydney Basin, Eastern Gondwana: Testing Pangea models and the timing of the end of the Kiaman Reverse Superchron

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ABSTRACT

Paleomagnetic and geochronologic data from the eastern margin of Gondwana have been obtained from the Gerringong Volcanics in the southern Sydney Basin, Australia. The corresponding paleomagnetic pole at 56.9°S, 154.8°E ($N = 131$; $A_{95} = 9.1°$) has a $^{40}Ar/^{39}Ar$ plagioclase plateau age of 265.05 ± 0.35 [0.46] Ma from the Bumbo Latite, and overlaps with recent radio-isotopic and paleomagnetic results published from Western Gondwana. The long-documented inconsistency between Middle Permian Eastern and Western Gondwanan paleomagnetic datasets is most likely an artefact of a lack of reliable paleomagnetic data from Eastern Gondwana for this period. A number of well-dated and recently published ca. 265 Ma paleomagnetic results from Gondwana and Laurussia are shown to be consistent with the Wegenerian Pangea A configuration, with a loose N-S fit of the continents for the Middle Permian. The lack of crustal overlap negates the need for a Pangea B configuration, which if valid must have been assembled to Pangea A by ca. 265 Ma.

The reverse polarity Bumbo Latite was sampled from the Kiaman type-section located in the southern Sydney Basin. Three cases of normal polarity were detected in the overlying Saddleback, Dapto, and Berkeley Latites, previously assigned to the Kiaman Reverse Superchron (KRS). We review KRS-aged magnetostratigraphic data and propose that an age assignment of 265 Ma most likely represents the termination of the non-reversing field, with longer stable intervals of normal polarity recorded and able to be correlated globally.

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1. Introduction

The Global Polarity Timescale (GPTS) contains the reversal record of Earth's magnetic field, and has been calibrated by correlating well-dated marine magnetic anomalies with magnetic reversals in sedimentary and igneous rocks for the past 160 Ma (Opdyke and Channell, 1996). Prior to the Jurassic, absence of oceanic anomalies requires classification of polarity intervals using global magnetostratigraphic correlation and relevant type sections (Opdyke and Channell, 1996). One such magnetostratigraphic section is from the Kiaman Reverse Superchron (KRS), preserved in Lower Carboniferous to Upper Permian volcanic rocks in eastern Australia (Cottrell et al., 2008; Irving, 1963; Irving and Parry, 1963; Opdyke and Channell, 1996).

The KRS is the interval from ca. 318 to 265 Ma when Earth's magnetic field remained fixed in a reverse polarity. With a duration of over 50 Ma, it represents the most prominent magnetostratigraphic interval of Paleozoic age, and signifies the longest lived stable geodynamo recorded (Irving and Parry, 1963; Opdyke and Channell, 1996; Steiner, 2006). Determining an age for the upper chronostratigraphic boundary will help improve models for the behaviour of Earth's magnetic field, and will provide a precise timing of geomagnetic phenomena to test potential correlations with significant Permian tectonic and biotic events.

Irving and Parry (1963) first classified the KRS from reversely magnetized rocks (Gerringong Volcanics) located in the southern Sydney Basin (Fig. 1), and noted that rocks of this age retained the same polarity as coeval rocks from other continents. Several hundred metres higher in the section a normal polarity was reported in the chocolate shales of the Narrabeen Group (Irving, 1963), and the inferred polarity transition was named the Illawara Reversal, because the end of the KRS was interpreted to lie within the underlying Illawarra Coal Measures (Fig. 2). The base of the KRS has been assigned an age of 318 Ma from two U/Pb ages

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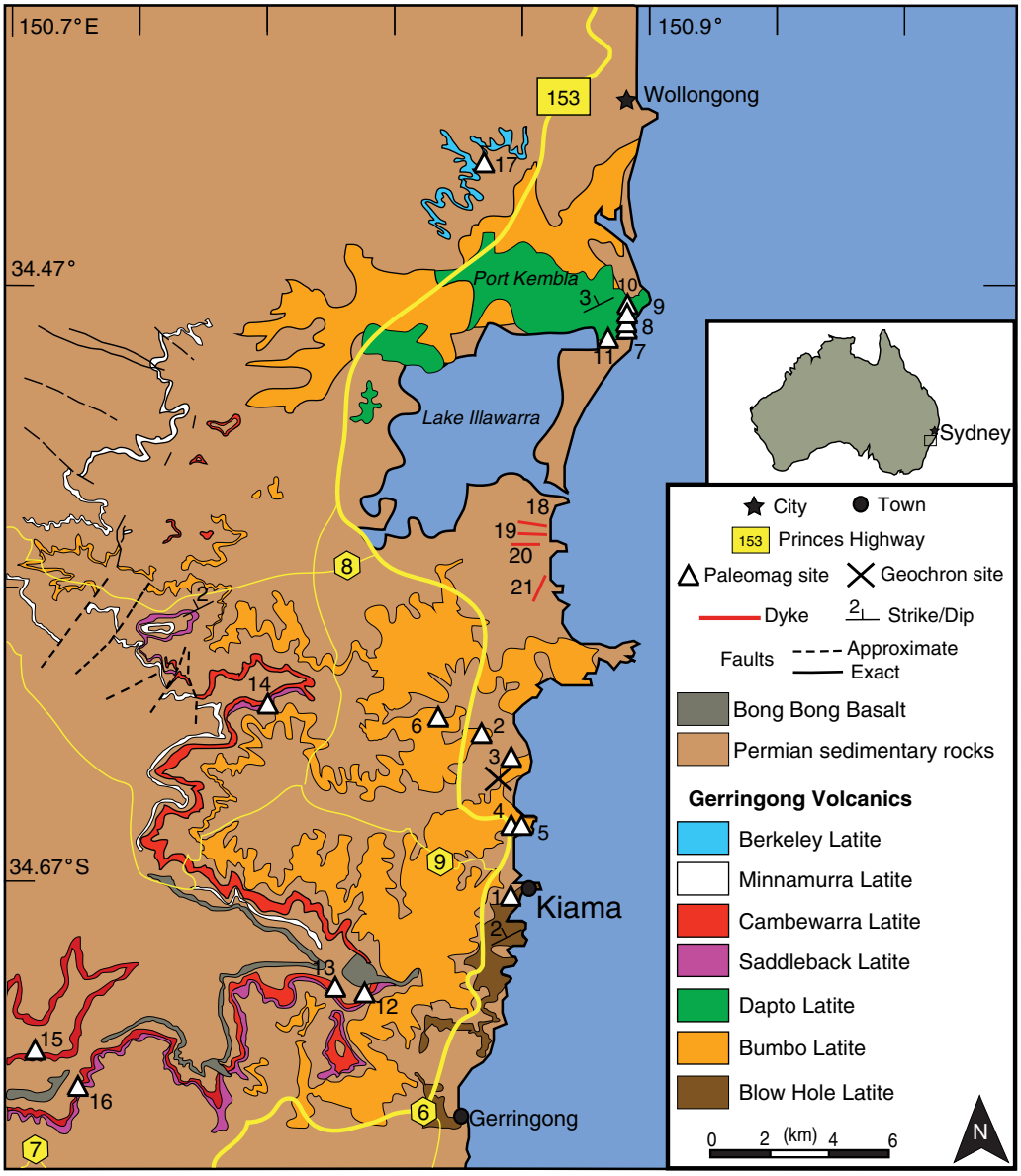


Fig. 1. Regional geologic map of the Kiama region, Sydney Basin, showing the distribution of Early –Late Permian volcanic and sedimentary rocks of the Shoalhaven Group and the overlying Illawarra Coal Measures (Bowman, 1974; Carr et al., 1999). See Fig. 2 for lithostratigraphic summary.

measured on zircon by SHRIMP of 317.8 ± 2.8 Ma (Eastons Arm Rhyolite) and 318.0 ± 3.4 Ma (Peri Rhyolite) in the northern Tamworth Belt of eastern Australia (Opdyke et al., 2000). These ages were determined using the SL13 standard zircon monitor that has been since shown to be heterogeneous in Pb/U (Black et al., 2003) so, although the ages appear coeval, heterogeneity in the standard means that the measured isotopic ratios cannot be corrected and are thus unreliable.

Early estimates from the Sydney Basin placed the base of the Illawara series at 263 Ma (Roberts et al., 1996), whereas more recent publications propose an older Wordian (~267 Ma; Steiner, 2006) or Roadian (~269 Ma; Lanci et al., 2013) age. The top of the KRS is commonly stated to be ~265 Ma in numerous publications and is based on a U/Pb (zircon) age of 265.4 ± 0.2 Ma from an ash bed located below the Capitanian type section in the Guadalupe Mountains, west Texas (Bowring et al., 1998; Menning et al., 2006). The corresponding magnetostratigraphic data from this section have not been published in full (Menning, 2000), and only reconnaissance-level paleomagnetic results from biostratigraphically correlative carbonate platform strata are available (Glenister et al., 1992; Peterson and Nairn, 1971).

During the Permian, the majority of Earth's continents were arranged in the supercontinent Pangea, which assembled during the Middle –Late Paleozoic, migrated northward at 8 cm/yr, and began to break up at ~180 Ma (e.g. Rogers and Santosh, 2003). The classical Wegenerian reconstruction of Pangea features closure of the Atlantic Ocean along the respective magnetic anomalies and there is abundant geological, paleontological, and geophysical evidence to support this configuration, commonly known as Pangea A (Bullard et al., 1965; Smith and Hallam, 1970). Paleomagnetic data were among the early datasets used to validate the hypothesis of plate tectonics, and the existing paleomagnetic data are consistent with pre-Late Triassic structural reconstructions. Although there is overwhelming evidence that the Atlantic opened from the Pangea A configuration, there remains an unresolved controversy regarding the Permian paleomagnetic reconstructions of Pangea (Aubele et al., 2012; Bachtadse et al., 2002; Briden et al., 1981; Domeier et al., 2011c, 2012; Irving, 1977; Muttoni et al., 2009, 2003; Torsvik et al., 2012; Van der Voo, 1993; Van der Voo and French, 1974; Van der Voo et al., 1984).

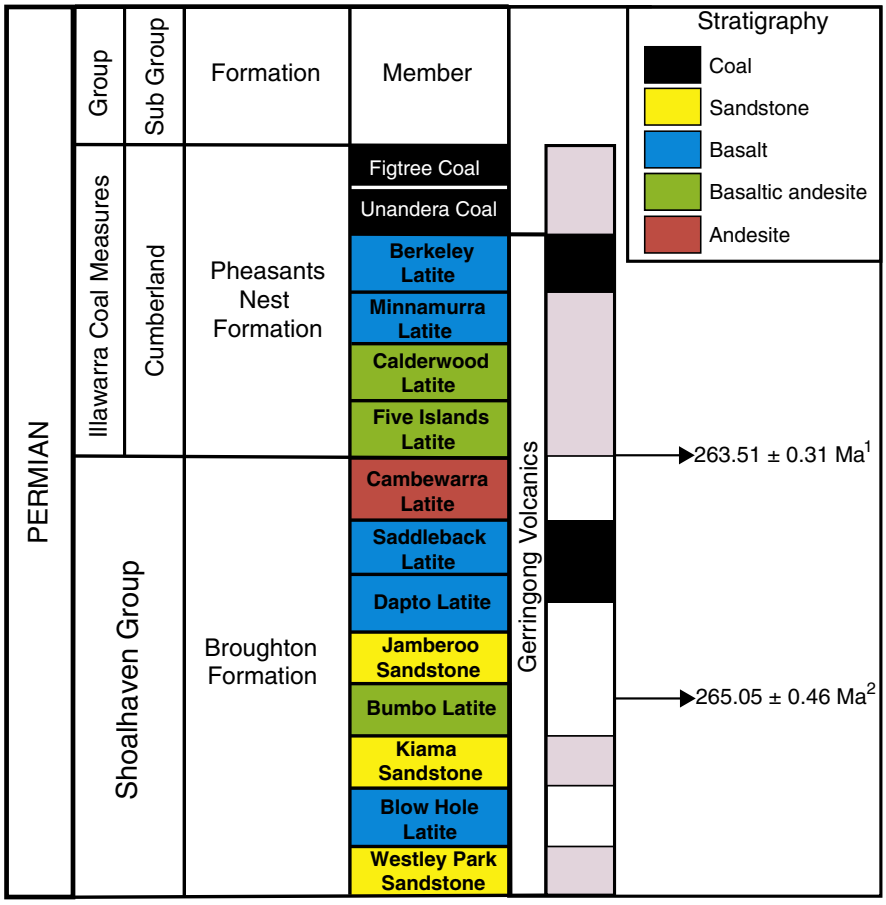


Fig. 2. Lithostratigraphy for the southern Sydney Basin (Campbell et al., 2001; Carr, 1983; Moffitt, 2000) showing the three sandstone members of the Broughton Formation, two coal members of the Pheasants Nest Formation, and igneous members of the Gerringong Volcanics. Note revised stratigraphic location of the Dapto and Saddleback Latites (P.F. Carr, pers. Comm.). Ref: 1 = Metcalfe et al., 2015; 2 = this study. Black = normal polarity; White = reverse polarity; Grey = Indeterminate.

The problem originates from the systematic longitudinal discrepancy between the Laurussian (North America, Eurasia) and Gondwanan (South America, Africa, India, and Australia) Apparent Polar Wander Paths (APWPs). If the continents had amalgamated into a single supercontinental landmass, then the two APWPs should coincide. However, in the Pangea A configuration, paleomagnetic poles from Gondwana plot ~20 –25° eastward of the reliable poles from Laurussia, resulting in a geologically unsupported latitudinal overlap between northern Gondwana and southern Laurussia (e.g. Van der Voo, 1993). This paleomagnetic discrepancy has led to an extensive debate regarding specific Early –Late Permian Pangean configurations. Irving (1977) proposed a configuration known as Pangea B, which applies a clockwise rotation (35°) of Gondwana such that the eastern coast of North America is placed against the north-western coast of South America (Fig. 3). This configuration was based on the available paleomagnetic data; however, it requires ~3000 km of dextral displacement between Gondwana and Laurussia in order for the Atlantic to open from the well documented Pangea A (Irving, 1977; Morel and Irving, 1981; Muttoni et al., 2003). The most recent paleomagnetic compilation in support of Pangea B shows transformation to Pangea A by an intra-Pangean dextral shear by the Late Permian –Early Triassic (Fig. 3), with post-Variscan transtension associated with widespread magmatism documented in the Southern Alps (Muttoni et al., 2009, 2003; Schaltegger and Brack, 2007). This compilation was restricted to results confined to a narrow band of paleolatitude, including the original study of the Gerringong Volcanics (Irving and Parry, 1963), and supports an Early –Middle Permian Pangea B configuration (Fig. 3; Muttoni et al., 2009).

Another hypothesis invokes non-dipole elements of Earth's magnetic field, or biases in the available paleomagnetic data —such as consequences of inclination shallowing or inclusion of low quality data (Domeier et al., 2012; Van der Voo, 1993). Inclination shallowing is a phenomenon which occurs in sedimentary rocks after the formation of a Detrital Remanent Magnetization (DRM) during post-depositional burial and compaction (Butler, 1992) that leads to systematic latitudinal bias in paleomagnetic poles (Anson and Kodama, 1987; Domeier et al., 2012; Kodama, 1997; Sun and Kodama, 1992; Tan et al., 2007; Tauxe, 2005; Tauxe and Kent, 2004; Tauxe et al., 2008). If non-dipole fields are considered, then the assumption that the Geocentric Axial Dipole (GAD) can be applied to Phanerozoic paleomagnetic data is no longer valid (Briden et al., 1971; Torsvik and Van der Voo, 2002; Van der Voo and Torsvik, 2001). Furthermore, global compilations of paleomagnetic data demonstrate that if Eastern Gondwana is excluded from APWP calculations, a better longitudinal fit is seen between the APWPs of Laurussia and Western Gondwana (Domeier et al., 2012; McElhinny and McFadden, 2000; Torsvik et al., 2012; Van der Voo, 1993).

Breakup of Gondwana did not begin until ~180 Ma (Rogers and Santosh, 2003), so reconciling the datasets between Eastern and Western Gondwana requires re-examination of the Late Paleozoic –Early Mesozoic paleomagnetic poles from Eastern Gondwana. Currently there are no well-dated results from this interval in Antarctica, India, and Madagascar, therefore, igneous rocks in Australia have the potential to provide insights into the problematic Eastern Gondwanan results (Domeier et al., 2012). The majority of available Early –Late Permian paleomagnetic results from Eastern Gondwana are from the New England Orogen (NEO) of eastern Australia (Clark, 1994; Irving and Parry, 1963; Klootwijk, 2003; Pisarevsky et al., 2016; Shaanan et al., 2015a). This orogen records several complex structural episodes of orocline formation and significant compression and folding associated with the

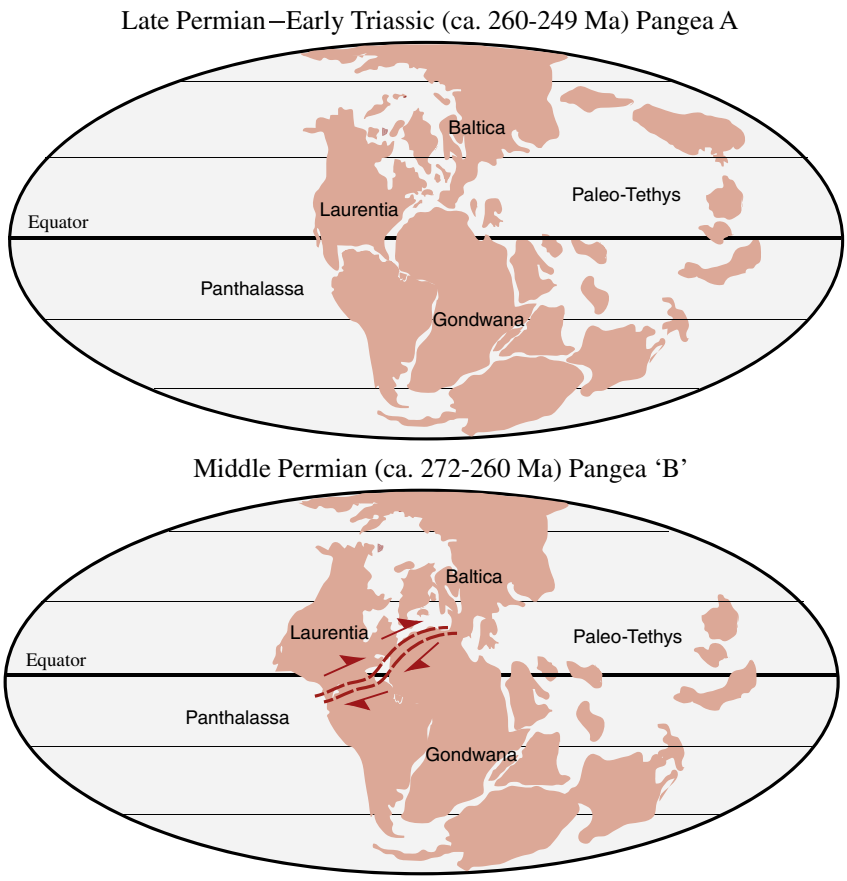


Fig. 3. Paleogeographic sketch of the Wegenerian Pangea 'A' configuration (valid from at least the Late Permian), and a proposed configuration for a Middle Permian Pangea 'B' with a ~3000 km dextral shear zone located between the continents of Laurussia and Gondwana (Muttoni et al., 2009).

Permian –Triassic Hunter Bowen Orogeny (McElhinny and McFadden, 2000; McElhinny et al., 2003; Pisarevsky et al., 2016; Van der Voo, 1993). Uncertainty regarding whether the NEO results were consistent with cratonic Gondwana, as well as the poor age control, led to the exclusion of Australian paleomagnetic results from 321 –183 Ma in the calculation of the Gondwana APWP (Torsvik et al., 2012).

This study presents new paleomagnetic and radio-isotopic results from the southern Sydney Basin (SSB), which represents the easternmost segment of Gondwana. The Gerringong Volcanics are a series of flat-lying basalts that were unaffected by the widespread compressional events to the north in the NEO (Fig. 4), and can be considered representative of cratonic Australia and Gondwana (Klootwijk, 2003; McElhinny and McFadden, 2000). This study will focus on the following three issues: 1) Can the paleomagnetic datasets of Eastern and Western Gondwana be reconciled for the Middle Permian through the analysis of well-dated igneous rocks? 2) Are the Middle Permian results consistent with a Pangea A or B configuration, and 3) what are the paleomagnetic implications for the timing of the end of the KRS?

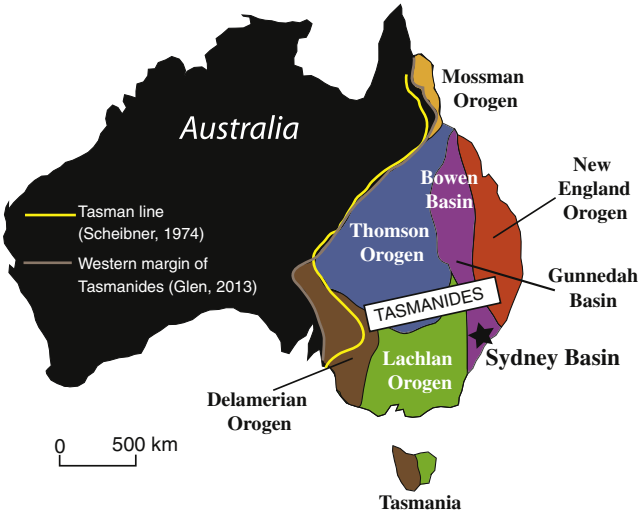


Fig. 4. Simplified terrane map of the Tasmanides of eastern Australia. The Sydney Basin is located south of the New England Orogen and the Gunnedah Basin (Modified from Glen, 2013).

2. Geologic background

The Tasmanides of Australia are located east of the four major Precambrian cratons (Yilgarn, Pilbara, North Australia, and Gawler), and consist of many displaced terranes that accreted to the Australian margin from the Cambrian to the Triassic (Glen, 2013; McElhinny and McFadden, 2000; Scheibner, 1974). Beginning in the Cambrian period, Australia formed the eastern extension of Gondwana, along with India, East Antarctica, Madagascar, South America, and Africa (Veevers, 2006). Westward subduction of the paleo-Pacific plate beneath the eastern margin of Australia resulted in several episodes of orogenesis from the latest Cambrian through the Early Triassic (Korsch et al., 2009b; Roberts et al., 2006). During the Early Permian, an eastern migration of the subduction zone initiated widespread crustal extension and resulted in the formation of the East Australian Rift System which extends from northern Queensland to southern New South Wales (Korsch et al., 2009a; Li and Rosenbaum, 2014; Roberts et al., 2006). The rift formed along the intersection of the western, Early to Middle Paleozoic Lachlan and Thomson Fold Belts, and the eastern,

Middle to Late Paleozoic NEO (Korsch et al., 2009a; Veevers and Powell, 1994). The Sydney Basin formed above the Late Devonian – Early Carboniferous Andean-type magmatic arc and forearc in the south, whereas the Bowen Basin developed behind the magmatic arc in the north (Veevers and Powell, 1994). During the Late Carboniferous to Early Permian, this magmatic arc was displaced through crustal transtension and the NEO was intruded by S-type granitoids and co-magmatic ignimbrites. The NEO was then uplifted during right-lateral shearing to form the initial stage of an orocline. By the Late Permian, the renewal of regional *E*-W compression, associated with the Hunter-Bowen Orogeny (~260 –230 Ma), further deformed the region, and the Tamworth Terrane was thrust onto the Sydney-Gunnedah-Bowen Basin (Harrington and Korsch, 1985; Holcombe et al., 1997; Korsch et al., 2009b; Landenberger et al., 1995; Li et al., 2012; McElhinny et al., 2003).

The SSB is located south of the NEO (Fig. 4), and consists of largely flat-lying and mostly undeformed sedimentary and igneous rocks (Fig. 1). The basin is dominated by alluvial fan, fluvial, and deltaic successions that prograded south-westward from the NEO as well as Permian mafic to intermediate lava flows (Gerringong Volcanics; Fig. 1). The Gerringong Volcanics consist of nine distinct basaltic to andesitic units interbedded with several volcaniclastic sandstones and tuffs (Carr, 1983; Fig. 2), and extend along the coast for over 150 km from Port Kembla in the north to Gerringong in the south (Bowman, 1974; Fig. 1). The base of the volcanic succession is the Broughton Formation (~370 m thick) of the Shoalhaven Group (Fig. 2), and consists of three basalts (Blow Hole, Dapto, Saddleback), one basaltic andesite (Bumbo), and one andesite (Cambewarra; Fig. 2; Carr, 1983; Bowman, 1974). The top of the volcanic succession belongs to the Pheasants Nest Formation of the Illawarra Coal Measures, and consists of two basaltic andesites (Five Islands, Calderwood) and two basalts (Minnamurra, Berkeley; Fig. 2; Carr, 1983). A tuff from the top of the Broughton Formation, which overlies the Cambewarra Latite, yielded a recent U/Pb ID-TIMS age on zircon of 263.51 ± 0.05 Ma [0.31] (Metcalfe et al., 2015), whereas the only data for the Berkeley Latite, located at the top of the Pheasants Nest Formation, is a K/Ar plagioclase age of 258 ± 5 Ma (Evernden and Richards, 1962).

The Gerringong Volcanics, as well as several shoshonitic intrusive complexes such as the Milton Monzonite (245 ± 6 Ma; K/Ar age; Facer and Carr, 1979) and various dyke emplacement events, have been interpreted as part of a larger magmatic episode within the SSB (Campbell et al., 2001; Carr, 1998; Carr et al., 1999; Facer and Carr, 1979; Moffitt, 2000; Schmidt and Embleton, 1981). The basalts were erupted from a volcanic arc associated with a west-dipping subduction zone about 40 km southeast of Kiama (Campbell et al., 2001). The arc was responsible for subaerial eruptions from a central vent as well as lateral eruptions from sub-aqueous vents that formed the Blow Hole and Bumbo Latites (Campbell et al., 2001). Detailed mapping shows thinning of the flows to the west and north, with associated feeder vents absent from onshore exposures (Campbell et al., 2001; Harper, 1915). Marine geophysical evidence traces the extrusive rocks offshore from Kiama for over 150 km to the northeast (Grybowski, 1992; Ringis et al., 1970), with paleoflow indicators supporting a northward advancement from the volcanic arc (Campbell et al., 2001).

The Blow Hole Latite is mid-grey in color with a fine-grained groundmass with plagioclase and zeolite clusters up to 2 mm in length and chlorite-coated augite phenocrysts. The Bumbo Latite is a porphyritic basalt with a mid-grey to black aphanitic groundmass and phenocrysts of plagioclase and minor clinopyroxene. Columnar jointing is observed in both units, and the Blow Hole Latite shows pillow structures in some areas, indicating a sub-aqueous eruption (Bowman, 1974; Campbell et al., 2001; Carr, 1998; Carr, 1983).

The mafic Dapto and Saddleback Latites are two geochemically similar units in the Gerringong Volcanics (Fig. 2), and have been thought to represent similar stratigraphic levels within the succession (Bowman, 1974; Carr, 1985, 1983). Both latites are dark in color, contain short prismatic pyroxene phenocrysts, and range from aphanitic to porphyritic in texture with a crystalline groundmass. There is no direct field evidence for continuity between these two units, with each unit thinning from its thickest exposure in the north near Port Kembla (Dapto Latite), and in the south at Saddleback Mountain (Saddleback Latite). Both underlie the Cambewarra Latite at these localities, however, up to 3 m of sedimentary rocks sit between the Dapto and Cambewarra Latites (Carr, 1983). In contrast, the contact between the Saddleback and overlying Cambewarra Latites lacks sedimentary deposits. This field relationship may suggest eruption of the Dapto Latite just prior to the Saddleback Latite, with Saddleback eruption in the south potentially simultaneous with post-Dapto sedimentary deposition in the north.

The Cambewarra Latite is the most felsic of the units, and consists of several flows interbedded with siltstone. It contains subparallel plagioclase phenocrysts in a felsic groundmass comprising iron oxide and feldspar microlites and interstitial iron oxide and chlorite (Bowman, 1974; Carr, 1998, 1985, 1983). The Berkeley Latite is more limited in exposure (Fig. 1) with more poorly developed columnar jointing compared to the Bumbo or Blow Hole Latites. It varies from aphanitic to porphyritic with plagioclase laths up to 10 mm long and pyroxene phenocrysts up to 5 mm across.

3. Methods

3.1. Paleomagnetism

Samples were collected in the field using a portable petrol-powered hand drill and oriented using a Brunton magnetic compass and a solar compass to correct for local magnetic declination. Twenty sites (8–10 samples per site) were sampled from the Gerringong Volcanics near Kiama, Australia, comprising the Blow Hole, Bumbo, Saddleback, Dapto, Cambewarra, Minnamurra, and Berkeley Latites (Fig. 1). Four dykes intruding the Jamberoo Sandstone were also sampled for paleomagnetic comparison and potential age determination (Fig. 1). Baked contact tests were conducted with 3–4 samples from the baked zone (~half-width of the dyke), 2–3 samples from the hybrid zone, and 3–4 unbaked samples within the host sandstone (Everitt and Clegg, 1962). A conglomerate test was conducted on the basaltic dropstones from the Blow Hole Latite in the overlying Kiama Sandstone to test for primary remanence of the lava flows (Watson, 1956). Fourteen dropstones ranging from 5 –80 cm in diameter were sampled at Shellharbour Beach (Fig. 1), including multiple samples per cobble to test for intra-cobble consistency.

Samples were cut into 10 cm^3 cylindrical cores, and measured at the Alpine Laboratory of Paleomagnetism in Peveragno, Italy, where a pilot study was conducted to determine the best treatment for isolation of the individual magnetic components. Samples were then stepwise demagnetized using either a 2G alternating field demagnetizer at fields up to 200 mT, or an ASC TD48 thermal demagnetizer with temperatures up to 650 °C. Samples were measured on a 2G DC-SQUID magnetometer, and paleomagnetic directions were isolated via principal component analysis (Kirschvink, 1980) using PaleoMag 3.1 (Jones, 2002). Individual components were calculated from three or more points, and directions were restricted to Maximum Angular Deviation (MAD) values ≤10° for lines and ≤15° for planes in calculation of the mean paleomagnetic pole. Linear-segment stable endpoints were combined with converging remagnetization circles and demagnetization planes in order to isolate the primary component of magnetization and to avoid the contaminating effects of strong overprint directions observed along the Cretaceous seaboard (Dunlop et al., 1997a; Dunlop et al., 1997b; McFadden and McElhinny, 1988; Schmidt and Embleton, 1981). Mean paleomagnetic directions and poles were calculated from the individual site means, and analysed using Fisher statistics (Fisher, 1953).

3.2. Rock magnetic fabric

In order to determine the magnetic fabric of the basalts and discern potential flow directions, 65 samples from the Dapto and Bumbo Latites were processed to determine the anisotropy of magnetic susceptibility (AMS). Principal axes of AMS were measured on a MFK1-FA Kappabridge using the 3-axis spinning protocol at the University of Western Australia. Results were plotted using the Anisoft 42® software (Chadima and Hrouda, 2009), and analysed using Jelínek statistics (Jelínek, 1981; Jelínek and Kropáček, 1978).

3.3. $^{40}Ar/^{39}Ar$ geochronology

Three exposures of the Bumbo and Dapto Latites were sampled for geochronology, and petrographic thin sections were made to examine the abundance and composition of the plagioclase crystals. One sample (B18) from the Bumbo Latite contains homogenous and unaltered plagioclase crystals suitable for mineral separation (Fig. 1; site 3). The sample was sawed, crushed, milled, and sieved using 150 to 200 μm meshes and then repeatedly washed and placed in an ultrasonic bath for 30 min. Non-magnetic plagioclase components were isolated using a Frantz Isodynamic Separator. Clean fractions of plagioclase were then handpicked under a binocular microscope avoiding any crystals containing impurities or that showed evidence of alteration or contamination.

Plagioclase separates (5.1 mg) were leached using diluted HF (2 N) for 5 min and rinsed repeatedly in distilled water. The remaining fraction was placed in a disc and irradiated for 40 h along with the FC standards (Jourdan and Renne, 2007) using an age of 28.294 Ma (±0.13%; Renne et al., 2011). The disc was then Cd-shielded (to reduce non-ideal nuclear interference reactions) and irradiated in the central position of the nuclear reactor at Oregon State University. A J-value of 0.01058060 ± 0.00000688 (±0.065%) was calculated from standard grains within the small pits, and mass discrimination was examined repeatedly during the analysis using an automated air pipette which gave a mean value of 0.992425 (±0.04%) per Dalton (atomic mass unit) relative to an air ratio of 298.56 ± 0.31 (Lee et al., 2006). Correction factors for interfering isotopes were ($^{39}Ar/^{37}Ar$) Ca = 7.6×10^{-4} (±1.2%), ($^{36}Ar/^{37}Ar$) Ca = 2.7×10^{-4} (±0.7%) and ($^{40}Ar/^{39}Ar$) K = 7.3×10^{-4} (±10%). The $^{40}Ar/^{39}Ar$ analyses were conducted at the Western Australian Argon Isotope Facility at Curtin University where the sample was step-heated using a continuous 100 W PhotonMachine™ CO_2 (IR, 10.4 μm) laser fired on the crystals for 60 s, with each of the standard crystals fused in a single step. The gas was purified in an extra low-volume stainless steel extraction line of 240 cm^3 and using two SAES AP10 (one operating at 450 °C and the other one at room temperature) and one GP50 getters. Argon isotopes were measured in static mode using a low volume (600 cm^3) ARGUS VI mass spectrometer from Thermofisher™ set with a permanent resolution of ~200 (Phillips and Matchan, 2013). Resulting measurements were conducted in multi-collection mode using four Faraday cup detectors to measure mass 40 to 37 and a 0-background compact discrete dynode ion counter to measure mass 36. The relative abundance of each mass was measured simultaneously using 10 cycles of peak-hopping and 33 s of integration time for each mass, and detectors were calibrated electronically and using air shot beam signals. The resulting data were then processed using the ArArCALC ® software (Koppers, 2002) and ages were calculated using the recommended decay constants (Renne et al., 2011). Relative abundances of Ar corrected from blanks, radioactive decay and fractionation are provided in the supplementary material. Criteria for the classification of a plateau are that: a) plateaus must incorporate at least 70% of ^{39}Ar; b) the plateau should be distributed over a minimum of three sequential steps that agree at the 95% confidence level; and c) satisfy a probability of fit (P) of at least 0.05. Plateau ages were calculated using the combined mean of plateau steps with each step weighted by the inverse variance of their individual analytical error, and reported at the 2σ level. All source of uncertainties are shown in brackets [±0.46 Ma].

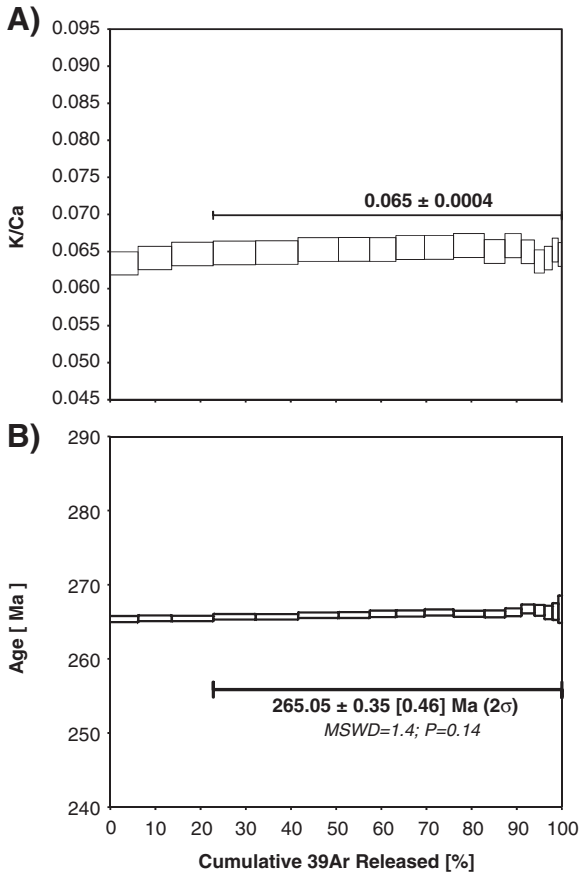


Fig. 5. $^{40}Ar/^{39}Ar$ results. (a) K/Ca ratios as a function of ^{39}Ar released during progressive stepwise heating (bars are plotted at 2σ error). (b) Age spectrum plot with a calculated plateau age of 265.05 ± 0.35 [0.46] Ma.

4. Results and interpretation

4.1. Geochronology

Sample B18 from the Bumbo Latite (Fig. 5) gave a consistent plateau age of 265.05 ± 0.35 [0.46] Ma, including 77% of the total ^{39}Ar released, where the uncertainty provided in square brackets includes all sources of uncertainties, which is appropriate when comparing with U/Pb data. The Mean Standard Weighted Deviation (MSWD) is 1.4 and the probability of fit (P) is 0.14. The K/Ca ratio is constant for the entire heating experiment, suggesting that the plagioclase separate was not affected by alteration (Verati and Jourdan, 2014) and thus, that the age represents the eruption age of the Bumbo Latite.

4.2. Paleomagnetism

4.2.1. Gerringong Volcanics

Seventeen out of twenty sites from the Gerringong Volcanics gave stable paleomagnetic directions suitable for analysis (Table 1). These include the Berkeley (one site), Cambewarra (two sites), Saddleback (three sites), Dapto (five sites), Bumbo (five sites), and Blow Hole (one site) Latites (Fig. 1). The majority of samples responded well to both thermal and alternating-field demagnetization. NRM intensity of the basalts ranges from 50 to 4000 mA/m with typical values in the range of 1000 –3000 mA/m. Progressive demagnetization of the samples to their respective unblocking temperatures and field values revealed three distinct paleomagnetic components (Fig. 6). Component

Table 1
Paleomagnetic results from the Gerringong Volcanics (ChRM).

Unit	Site	S_{lat} (°S)	S_{long} (°E)	N	P	*D* (°)	*I* (°)	α_{95}	*k*	P_{lat} (°S)	P_{long} (°E)
Blow hole	1	34.672	150.858	7	R	226.0	82.6	4.6	224.4	44.0	136.3
Bumbo	2	34.621	150.845	6	R	199.1	78.9	9.8	61.5	54.4	139.0
Bumbo	3	34.631	150.850	9	R	122.4	72.9	6.5	79.9	45.6	190.1
Bumbo	4	34.651	150.853	5	R	212.2	86.6	6.1	230.4	40.3	146.1
Bumbo	5	34.653	150.856	10	R	211.7	75.7	3.8	162.0	55.5	125.9
Bumbo	6	34.616	150.828	9	R	208.5	77.8	4.2	174.0	53.9	132.1
Dapto	7	34.477	150.914	5	R	130.8	77.3	13.7	59.0	47.5	178.3
Dapto	8	34.477	150.914	6	R	145.3	66.9	12.4	100.6	60.5	199.6
Dapto	9	34.477	150.914	7	R	273.3	85.6	9.5	95.2	33.5	140.4
Dapto	10	34.477	150.914	10	N	315.7	−79.2	6.3	146.9	47.7	172.6
Dapto	11	34.491	150.914	9	N	42.5	−70.2	19.4	19.2	54.6	108.0
Saddleback	12	34.691	150.823	9	N	333.4	−68.3	11.9	37.0	64.6	191.3
Saddleback	13	34.698	150.824	7	R	182.7	70.3	3.2	350.0	70.2	146.2
Saddleback	14	34.610	150.756	8	N	8.8	−71.9	14.3	20.6	67.0	138.4
Cambewarra	15	34.726	150.654	7	R	142.0	57.9	10.7	45.0	59.5	222.1
Cambewarra	16	34.732	150.687	6	R	205.2	70.2	10.1	45.0	63.7	116.5
Berkeley	17	34.437	150.837	10	N	315.5	−83.3	17.5	17.5	43.2	163.5

Notes: S_{lat} = site latitude, S_{lon} = site longitude, N = number of samples used for calculation of the mean direction, P = polarity, D = declination, I = inclination, α_{95} = radius of the cone of 95% confidence about the mean direction, *k* = precision parameter (Fisher, 1953), P_{lat} = pole latitude, P_{long} = pole longitude.

A is a low temperature overprint magnetization stable to about 250 – 400 °C (20 mT) and is present in nearly every sample (Fig. 6). Several overprint directions and thermal/alteration events have been reported in eastern Australia, including Middle –Late Cenozoic low temperature oxidation and weathering (Klootwijk, 2003, 2002; McQueen, 2002; Nott et al., 1991; Pillans, 2002; Pisarevsky et al., 2016; Schmidt and Williams, 2002), and a possible thermo-viscous overprint associated with the opening of the Tasman Sea (Dunlop et al., 1997a, 1997b; Klootwijk, 2002, 1985; Lackie and Schmidt, 1996; Schmidt and Embleton, 1981). Component A has a mean direction of Dec = 345.0°, Inc. = −62.6° (α_{95} = 10.1°; Tables 2 and 3; Fig. 6), and is shallower and closer to the present day field direction than results which have been attributed to extended Cretaceous burial and uplift (Dunlop et al., 1997a; Dunlop et al., 1997b; Schmidt and Embleton, 1981), and is most likely a Viscous Remanent Magnetization (VRM).

Component B_R begins to unblock around 250 –400 °C (20 mT) and is stable to about 560 °C (40 mT) in the Bumbo Latite (Fig. 6b), 640 °C (200 mT) in the Blow Hole Latite, 560 °C (80 mT) in the Dapto Latite (Fig. 6d), and to about 650 °C (200 mT) in the Cambewarra Latite (Fig. 6a). This component has a southerly declination and a steep-down direction (D = 172.6°, *I* = 78.1°, α_{95} = 6.7°, *N* = 84; Table 3; Fig. 6) that has been commonly assigned to the KRS (Cottrell et al., 2008; Irving and Parry, 1963). The Saddleback, Dapto, and Berkeley Latites preserve three magnetic components. The first is the same low temperature Component A, the second is a reverse and steep-down direction stable from about 400 –540 °C named Component C (D = 239.0°, *I* = 68.1°, α_{95} = 28.0°, *N* = 16), and the third is a northerly and steep-up direction (Component B_N) that is present in both thermal and alternating field demagnetization (Fig. 6). This direction (B_N; D = 354.7°, *I* = −77.0°, α_{95} = 10.7°, *N* = 46; Table 3; Fig. 6) begins to unblock around 450–475 °C and is stable to about 560 °C in the Dapto Latite (Fig. 6c), 565–640 °C (50–100 mT) in the Saddleback Latite (Fig. 6e-f), and ranges from 475–550 °C to 565–630 °C in the Berkeley Latite (Fig. 6g-h). Component C (present from 400 –540 °C) is shallower in inclination (~68°) than the bipolar components B_N and B_R (~78°) and in some samples has completely remagnetized the ChRM direction, especially in the more altered and eroded exposures such as the Berkeley Latite (Fig. 1).The range in unblocking temperatures of B_R and B_N suggests that the ChRM is held by a mixture of magnetite and hematite (Fig. 6), which agrees with prior rock magnetic analyses of the Dapto and Bumbo Latites (Cottrell et al., 2008). Components A and B_N are paleomagnetically distinct at the 95% confidence level, while Components B_R and B_N are statistically antipodal and pass a reversal test with a classification of R_a (γ_o = 2.2°, γ_c = 5.8°; McFadden and McElhinny, 1990; Fig. 7). A reversal test restricted to stable endpoint directions gives a classification of R_b (γ_o = 8.65°, γ_c = 9.98°; McFadden and McElhinny, 1990).

4.2.2. Magnetostratigraphy

At the Port Kembla locality (Fig. 1), two basalt flows of the Dapto Latite were sampled (5 sites/40 samples/4 stratigraphic levels). The lower flow is characterized by a green porphyritic basalt with large plagioclase laths and secondary calcite-filled amygdales, whereas the upper flow is darker in color and contains elongate plagioclase crystals aligned layer parallel. The lower flow contains both Components A and B_R, with stable endpoints and great circle demagnetization trajectories converging at a common intersection point to the south and steep-down (B_R; Fig. 6d). The upper flow shows stable endpoints and converging demagnetization trajectories to the north and steep-up over 500 °C (Component B_N; Fig. 6c). Great circle and stable endpoint directions were combined using the iterative procedure of McFadden and McElhinny (1988), which nullifies the issues with subparallel great circles in determination of the true ChRM direction. The two flows pass the reversal test with a classification of R_b (γ_o = 6.4°, γ_c = 10.4°; McFadden and McElhinny, 1990).

The upper Dapto and Saddleback Latites (normal polarity) overlie the thick KRS Bumbo (265.05 ± 0.35 Ma) and Blow Hole Latites (reverse polarity), and so an age of ca. 265 Ma can be directly correlated with the first post-KRS reversal (Illawarra Reversal; Fig. 8). The Berkeley Latite can be tentatively linked with the early Capitanian reversals of the GPTS (Fig. 8). An age of 263.51 ± 0.05 [0.31] Ma from the uppermost Broughton Formation (Metcalfe et al., 2015) and the $^{40}Ar/^{39}Ar$ age of 265.05 ± 0.35 [0.46] from the lower Broughton Formation, as well as a positive reversal test, suggests that the Gerringong Volcanics span a statistically sufficient amount of geologic time to average secular variation. This is further corroborated by a VGP angular dispersion value of S = 20.0° (K = 16.3) from 17 sites, which is consistent with the calculated paleolatitude of 67° (Butler, 1992). Virtual Geomagnetic Poles (VGPs) were calculated from the individual site means (12 Reverse, 5 Normal) and were combined to give a mean paleomagnetic pole at 56.9°S, 154.8°E (A_{95} = 9.1°, *N* = 17; Table 3).

4.2.3. Conglomerate test

Results from the volcanic dropstones at Shellharbour yield a negative conglomerate test (*R* = 6.7, R_0 = 4.7, R > R_0 non-random distribution; Watson, 1956) with a mean direction of Dec = 357.5°, Inc. = −69.4°, and α_{95} = 28.1° (Table 4; Fig. 9a). Individual cobbles show internal consistency (3–4 samples per clast) with a randomly oriented overprint direction and a secondary high temperature

direction (Fig. 9b). This high temperature direction is not statistically different from Component A (Fig. 9a), and also likely reflects a VRM.

Although the conglomerate test cannot directly confirm a primary magnetization for the dual-polarity (Components B_N and B_R) Gerringong Volcanics, the positive reversal test observed between these two high temperature directions suggests that the ChRM has been adequately filtered from secondary magnetizations (Figs. 6 and 7).

4.2.4. Shellharbour dykes

Four out of the five dykes (D1–D5; Table 4) sampled gave stable paleomagnetic directions suitable for analysis (Fig. 1). The dykes range from grey-green vesicular basalt to mid-grey-green aphanitic basalt showing ferromagnesian phenocrysts (augite) up to 2 mm in length (Bowman, 1974). One dyke at Barrack Point (D3) is coarse grained and black in color, with indurated edges. Typical dyke widths ranged from 0.7 –1.5 m, with either a NE (35°; D5) or *E*-SE (90 –110°; D1, D2,

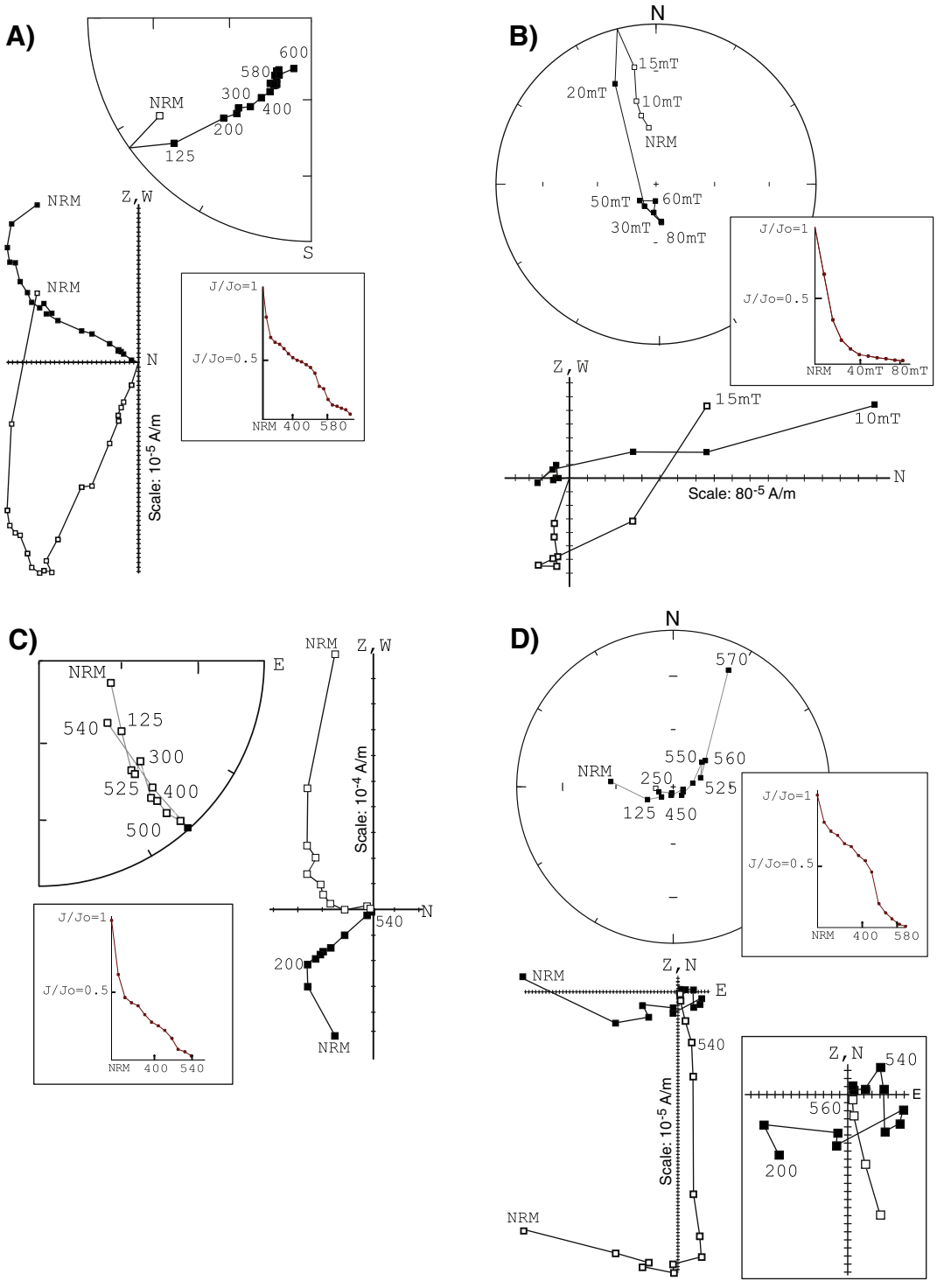


Fig. 6. Representative paleomagnetic demagnetization diagrams (equal area diagrams, orthogonal vector plots, and J/J0) for the Gerringong Volcanics showing the dual-polarity ChRM and overprint directions. A) Cambewarra Latite, B) Bumbo Latite, C–D) Dapto Latite. Orthogonal insets show the final demagnetization steps. Solid (open) squares represent projections on the horizontal (vertical) plane in the orthogonal plots while up (down) pointing paleomagnetic directions are indicated by open (closed) squares in the stereoplots. *E*–F) Saddleback Latite, G) Blow Hole Latite, H) Berkeley Latite.

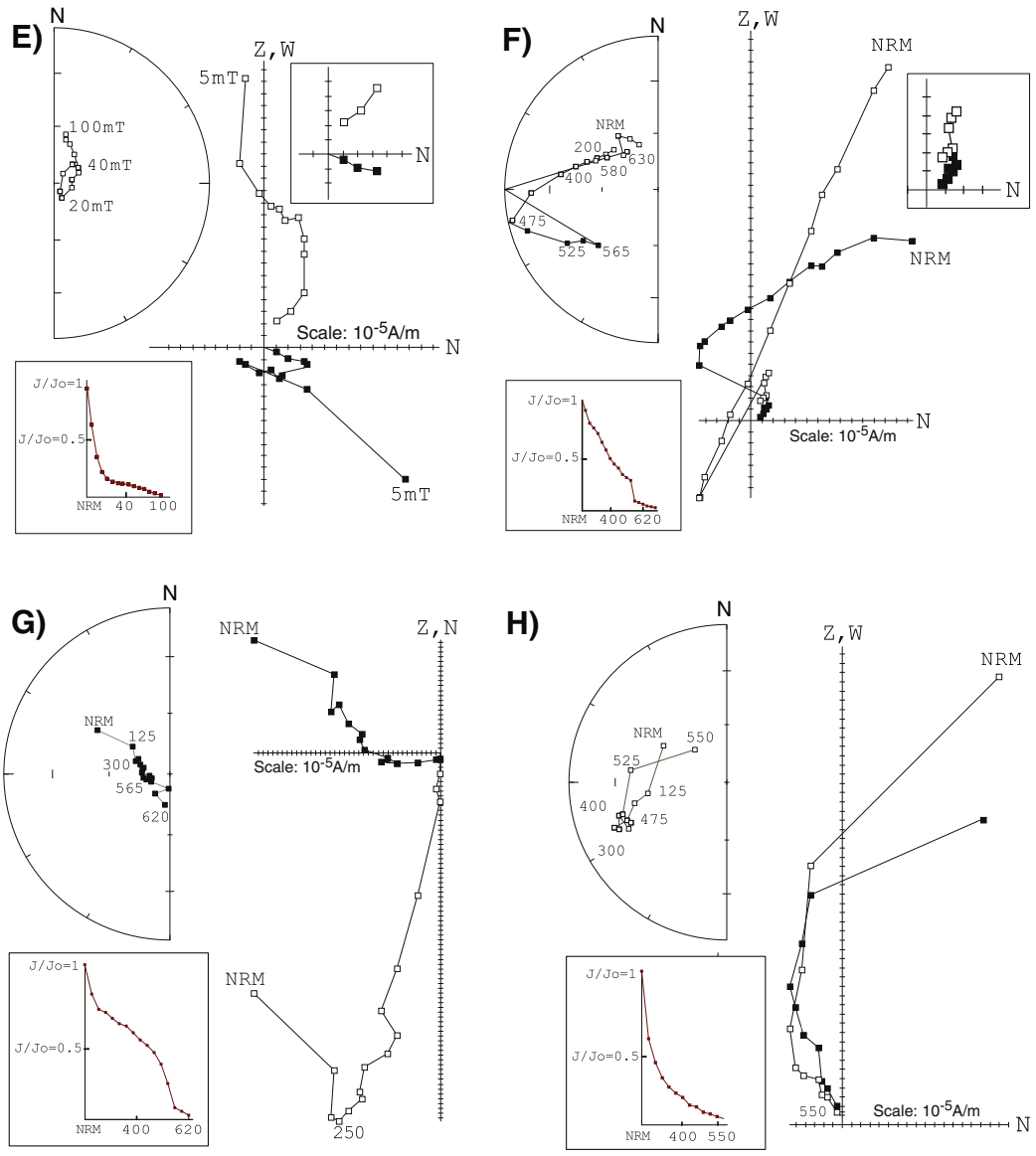


Fig. 6 (*continued*).

and D3) trend. One dyke at Shellharbour is highly altered to yellow-brown clays, and did not yield coherent results during demagnetization (D4). The remaining dykes show two distinct remanence components during progressive demagnetization. Component D_A is a low temperature normal polarity overprint direction present to 250 –350 °C (Table 4). This component is not statistically different from Component A in the Gerringong Volcanics and the volcanic dropstones (Fig. 9a). Two of the dykes (D1 and D3) responded well to thermal demagnetization and revealed a second normal component of magnetization over 350 °C, with steepening inclinations during progressive demagnetization (Component D_N). D2 and D5 showed better removal of overprint directions using alternating field demagnetization, and revealed a likely primary reverse direction named Component D_R. This component is not statistically different from Component C in the Gerringong Volcanics (Fig. 10). The four dykes pass a reversal test ($\gamma_o = 16.0°$, $\gamma_c = 39.3°$; McFadden and McElhinny, 1990).

Three dual-polarity dykes (two reverse, one normal) reported from the Newcastle Coal Measures (Sakai et al., 2005) have two ages of 79.6 ± 1.8 Ma (K/Ar, plagioclase) and 78.0 ± 2.2 Ma (K/Ar, whole rock), and have been correlated with the Cretaceous rifting of the Tasman Sea. Unfortunately, the accuracy of K/Ar results on such old rocks is usually questionable due to the inability of the technique to identify alteration (e.g., Verati and Jourdan, 2014). The mean paleomagnetic direction for the Cretaceous dykes (Sakai et al., 2005) is similar in inclination (68 –73°) to the two reverse polarity dykes reported here, and plots within the α_{95} confidence cone of both the mean dyke direction and Component C (Fig. 10). Interestingly, the α_{95} cone of Component C also overlaps with the original study of the Gerringong Volcanics (Fig. 10; Irving and Parry, 1963). It is likely that the dykes sampled here and by Sakai et al. (2005) are a magmatic product of the Cretaceous rifting event which also produced the remagnetization (Component C) that affected prior paleomagnetic investigations in the Sydney Basin (Dunlop et al., 1997a, 1997b; Irving and Parry, 1963; Schmidt and Embleton, 1981). Since the dykes have been shown to be dual-polarity (Sakai et al., 2005), it is likely that some rift-related magmatism post-dated the end of the Cretaceous Normal Superchron (121 –83 Ma), which is broadly consistent with the reported K/Ar ages (~79 Ma).

Baked contact tests in the host sandstone support a primary magnetization for the two high temperature (field) dyke directions (Fig. 11; Table 4). The combined unbaked direction is not statistically different from Components A and D_A, and is likely also a VRM. The positive baked contact test is demonstrated by D2 (Fig. 11), which gives a

Table 2
Paleomagnetic results from the Gerringong Volcanics (overprints).

Unit	Site	S_{lat} (°S)	S_{long} (°E)	N	P	D (°)	I (°)	α_{95}	k
Component A									
Blow hole	1	34.672	150.858	4	N	341.5	−65.6	16.6	23.6
Bumbo	2	34.621	150.845	5	N	335.1	−62.0	21.7	10.7
Bumbo	3	34.631	150.850	4	N	278.9	−68.8	72.6	1.8
Bumbo	4	34.651	150.853	9	N	15.8	−63.1	8.9	30.6
Bumbo	5	34.653	150.856	11	N	2.6	−32.9	17.7	6.9
Bumbo	6	34.616	150.828	7	N	354.4	−49.0	14.5	15.6
Dapto	7	34.477	150.914	5	N	12.7	−48.3	24.2	8.7
Dapto	8	34.477	150.914	5	N	73.6	−61.5	72.9	1.6
Dapto	9	34.477	150.914	5	N	346.9	−25.8	32.9	5.1
Dapto	10	34.477	150.914	8	N	322.4	−51.9	33.9	3.1
Dapto	11	34.491	150.914	6	N	127.8	−82.4	35.5	3.6
Saddleback	12	34.691	150.823	8	N	320.9	−62.8	14.9	12.9
Saddleback	13	34.698	150.824	6	N	332.7	−58.0	5.3	135.5
Saddleback	14	34.610	150.756	6	N	4.8	−69.3	17.3	13.3
Cambewarra	15	34.726	150.654	5	N	321.8	−48.6	10.1	46.3
Cambewarra	16	34.732	150.687	4	N	294.5	−73.0	17.5	21.3
Berkeley	17	34.437	150.837	12	N	324.6	−65.1	12.5	12.0
Component C									
Dapto	10	34.477	150.914	7	R	129.4	85.0	4.7	143.6
Dapto	11	34.491	150.914	7	R	184.9	48.7	29.0	4.3
Saddleback	12	34.691	150.823	7	R	260.7	65.1	24.0	6.2
Saddleback	14	34.610	150.756	6	R	228.8	53.1	27.0	5.9
Berkeley	17	34.437	150.837	4	R	297.3	48.1	37.3	5.3

Notes: S_{lat} = site latitude, S_{lon} = site longitude, N = number of samples used for calculation of the mean direction, P = polarity, D = declination, I = inclination, α_{95} = radius of the cone of 95% confidence about the mean direction, k = precision parameter (Fisher, 1953).

primary reverse direction (Component D_R) of Dec = 155.8° and Inc. = 67.8° (α_{95} = 7.3°, Table 4) at high fields (>30 mT). Seven centimetres from the contact, the host sandstone shows a progressive unblocking of Component D_A towards the mean dyke direction (Component D_R; Dec = 84.7°, Inc. = 70.3°, α_{95} = 5.5°) within the baked zone (Fig. 11b). Ten centimetres from the contact the host sandstone shows partial remagnetization, with an intermediate (hybrid) direction between Components D_A and D_R (Fig. 11c). Distant unbaked samples show no evidence for the steep reverse polarity Component D_R, and yield directions restricted to Component D_A (Fig. 11d). The dykes sampled in this study help confirm potential sources for overprint directions and information regarding specific magmatic events in the SSB; however, it is noted that sampling of the dykes has been limited to four sites in this study and only three sites by Sakai et al. (2005). Further paleomagnetic and geochronologic work is required to establish representative paleomagnetic results for tectonic and APWP considerations. The presence of both normal and reverse polarities in the Gerringong Volcanics within the unblocking temperatures of both magnetite (500–570 °C) and hematite (580–650 °C), the careful resolution and classification of individual overprint components through the appropriate field tests, and detailed rock magnetic work (Cottrell et al., 2008) support a magnetostratigraphic interpretation for the paleomagnetic results, as opposed to laboratory induced self-reversal of ilmeno-hematite. The results reported here pass a reversal test, and fall on the correct segment of the Gondwana APWP, which suggests that the pole has been adequately filtered of secondary and viscous remanent magnetizations.

The following stratigraphic and magmatic history is based on the available paleomagnetic and age constraints: initial eruption of the Gerringong Volcanics (single polarity; Component B_R) at ~270 Ma, deposition of dropstones of the Blow Hole Latite, continued eruption of basalts until ~260 Ma (dual polarity; Components B_N and B_R), intrusion of the Milton Monzonite at a later, yet unconstrained stage, remagnetization related to Cretaceous magmatism (Component C/D_R), and finally widespread acquisition of a VRM (Component A/D_A). The mean direction reported from the original study of the Gerringong Volcanics (Irving and Parry, 1963) does not show statistical agreement with the mean ChRM reported here (Component B_N/B_R); however, it plots within the α_{95} confidence circle of Component C (Fig. 10). It is very likely that this overprint direction as well as other remagnetization events have affected previous results from the Sydney Basin due to the inadequate (partial) demagnetization methods common in older paleomagnetic studies. It is, therefore, necessary to revisit additional paleomagnetic studies from Eastern Gondwana in order to resolve the potential tectonic issues associated with these results, as well as for polarity considerations.

4.3. Magnetic fabric

AMS results from the Bumbo and Dapto Latites are presented in Table 5 and Fig. 12. Field observations of the Bumbo Latite such as elongation of vesicles, overstepping and thinning of flow units to the north and west, orientation of lava tubes, and the entrainment of sediment within flowing lava suggest a northern flow direction (Campbell et al., 2001). AMS results reported here also confirm a general northward flow direction (Fig. 12), which is compatible with the proposed volcanic source located some tens of kilometres southeast of Kiama (Campbell et al., 2001; Grybowski, 1992; Ringis et al., 1970). The measured anisotropy factors did not show any substantial intra-site variation for the Bumbo and Dapto Latites (Table 5; Fig. 12). The shape of the AMS ellipsoid from the Bumbo Latite (Fig. 12) is generally oblate with a shape factor (T) = 0.243 (Table 5), while Jelinek diagrams from the Dapto Latite are not conclusive with negative T values implying a more prolate shape for the ellipsoid (Table 5; Fig. 12). The similarity between the shape factors (T) and difference shape factors (U) for each unit suggests a very low degree of anisotropy (Jelínek, 1981), which confirms that the volcanics rocks are largely undeformed and suitable for paleomagnetism.

5. Discussion

5.1. The Gondwanan APWP

The problems with Gondwanan paleomagnetic data from 290 – 230 Ma with respect to the paleomagnetic poles of Laurussia have

Table 3
Mean directions for the Gerringong Volcanics.

Component	S_{lat} (°S)	S_{long} (°E)	B/N	P	D (°)	I (°)	α_{95}	k	P_{lat} (°S)	P_{long} (°E)	A_{95}
Mean normal (B_N)			5/46	N	354.7	−77.0	10.7	51.7			
Mean Reverse (B_R)			12/84	R	172.6	78.1	6.7	43.5			
Grand Mean(B_N + B_R)			**17/131**	**R**	**173.3**	**77.8**	**5.2**	**48.3**	**56.9**	**154.8**	**9.1**
*Mean (B_N + B_R)**			*14/61*	*R*	*180.1*	*76.4*	*3.2*	*33.7*	*60.4*	*150.7*	*5.7*
Irving and Parry (1963)	34.600	150.800	11/43	R	232.0	81.0	6.0	26.0	44.0	132.0	11.4
Component C			5/31	R	239.0	68.1	28.0	8.4			
Component A			17/114	N	345.0	−62.6	10.1	13.5			

Notes: S_{lat} = site latitude, S_{lon} = site longitude, B = number of sites, N = number of samples used for calculation of the mean direction, P = polarity, D = declination, I = inclination, α_{95} = cone of 95% confidence about the mean direction, k = precision parameter (Fisher, 1953), P_{lat} = pole latitude, P_{long} = pole longitude, A_{95} = radius of the 95% confidence circle about the calculated mean pole, * = Mean excluding great circles, Bold = recommended paleomagnetic mean.

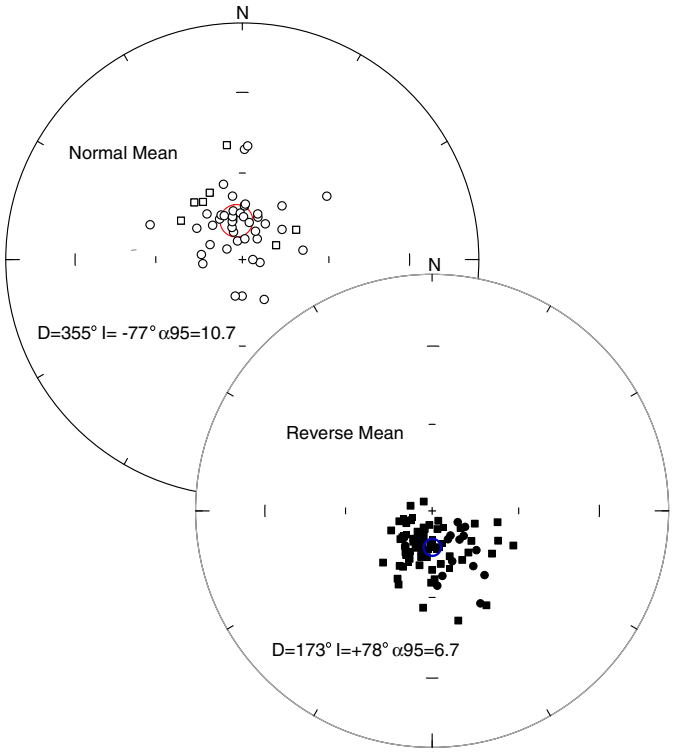


Fig. 7. Equal area diagrams displaying the mean Normal and Reverse ChRM directions from the Gerringong Volcanics. Up (down) pointing paleomagnetic directions are indicated by open (closed) squares in the stereoplots. Arc-constraints for samples that do not reach stable endpoints are calculated with the method of McFadden and McElhinny (1988), and are shown with circle symbols. Red and blue ovals represent the cone of 95% confidence about the mean direction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

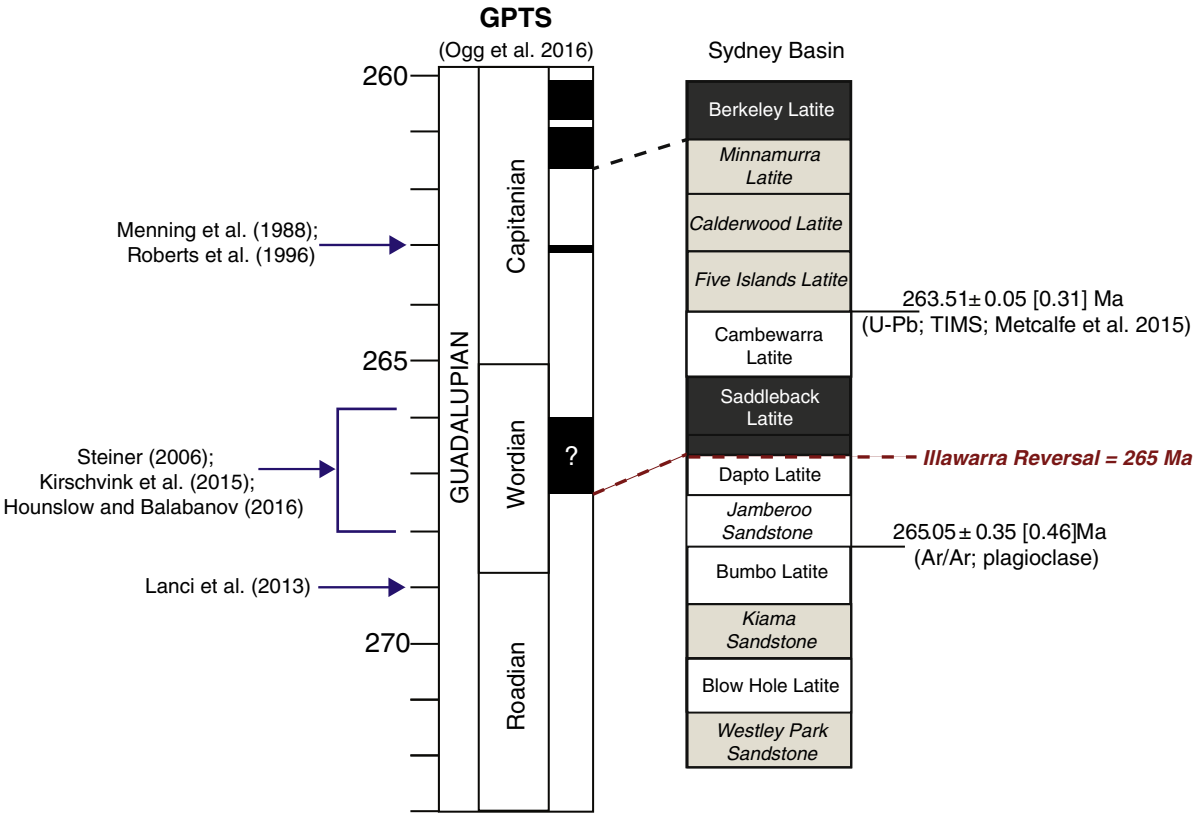


Fig. 8. Stratigraphic estimate for the uppermost KRS at 265 Ma, observing the change of polarity from the Blow Hole and Bumbo (Reverse) to Dapto (Mixed) and Saddleback (Normal) Latites, located one unit below the base of the Illawarra Coal Measures. Results are compared with the most recent Global Polarity Timescale (Ogg et al., 2016) and prior estimates of the duration of the KRS.

Table 4
Paleomagnetic field tests.

Map	Site	S_{lat} (°S)	S_{long} (°E)	B/N	P	*D* (°)	*I* (°)	α_{95}	*k*	P_{lat} (°S)	P_{long} (°E)	A_{95}
18	D1	34.562	150.874	10	N	344.2	−79.6	4.3	146.6	53.7	160.0	
	Baked	*34.562*	*150.874*	*2*	*N*	*316.5*	*−78.6*	*12.0*	*436.0*			
	Unbaked	*34.562*	*150.874*	*3*	*N*	*353.8*	*−60.7*	*13.6*	*83.5*			
19	D2	34.563	150.872	12	R	155.8	67.8	7.3	85.8	66.1	190.7	
	Baked	*34.563*	*150.872*	*2*	*R*	*100.3*	*73.7*	*29.7*	*72.6*			
	Unbaked	*34.563*	*150.872*	*4*	*N*	*355.0*	*−62.5*	*7.8*	*139.7*			
20	D3	34.564	150.872	13	N	9.2	−80.1	3.7	261.3	53.5	145.8	
	Baked	*34.564*	*150.872*	*3*	*N*	*32.5*	*−80.4*	*4.9*	*642.0*			
	Unbaked	*34.564*	*150.872*	*3*	*N*	*355.4*	*−66.2*	*11.0*	*127.3*			
21	D5	34.581	150.867	11	R	190.8	73.9	6.5	119.9	63.6	138.7	
	Normal Dyke Mean (D_N)			2/21	N	2.7	79.3	3.4	90.3			
	Reverse Dyke Mean (D_R)			2/23	R	225.9	83.6	25.7	3.2			
	Sakai et al., 2005			*2*	*R*	*228.9*	*66.6*	*6.4*				
				1	*N*	*48.9*	*−66.6*	*6.4*				
	Component D_A			4/19	N	0.5	−64.6	14.5	6.4			
	Unbaked Mean			3/10	N	354.7	−63.1	4.2	131.4			
	Conglomerate Mean	34.578	150.871	14/14	N	357.5	−69.4	28.1	4.2	71.4	155.6	44.0

Notes: S_{lat} = site latitude, S_{lon} = site longitude, B = number of sites, N = number of samples used for calculation of the mean direction, P = polarity, D = declination, I = inclination, α_{95} = cone of 95% confidence about the mean direction, *k* = precision parameter (Fisher, 1953), P_{lat} = pole latitude, P_{long} = pole longitude, A_{95} = radius of the 95% confidence circle about the calculated mean pole. Baked contact test = positive, conglomerate test = negative.

been well documented (Domeier et al., 2012; McElhinny and McFadden, 2000; Torsvik et al., 2012; Van der Voo, 1993). Since initiation of rifting in Gondwana did not begin until ~180 Ma (Rogers and Santosh, 2003), a potential source of contamination might be present within Eastern Gondwanan results for this time period.

Paleomagnetic poles from igneous rocks (and sedimentary rocks corrected for inclination shallowing) within the 270 –260 Ma time window have been selected for comparison with the Gerringong Volcanics at 265 Ma, with the recommended Gondwana APWP (Torsvik et al., 2012) plotted for reference only (Fig. 13). Each pole presented here is evaluated using the reliability criteria of Van der Voo (1990), which include: 1: Well determined rock age and a determination that the magnetization is of the same age; 2: Sufficient number of samples ($N > 24$), $K \geq 10$ and $A_{95} \leq 16°$; 3: Adequate demagnetization that demonstrably includes vector subtraction; 4: Field tests that constrain the age of magnetization; 5: Structural control and tectonic coherence with craton or block involved; 6: The presence of reversals; 7: No resemblance to paleopoles of younger age by more than a period (not necessarily remagnetization). Criteria met are assigned a value of 1 while unmet criteria are marked as 0 (Tables 6 and 7).

5.2. Western Gondwana (270 –260 Ma)

Selection of Middle Permian paleomagnetic poles from Western Gondwana (Table 6) were restricted to results with $Q \geq 5$ (Van der Voo, 1990) and include the Sierra Chica, La Pampa (Domeier et al., 2011b) and Upper Choiyoi Group of central Argentina (Domeier et al., 2011c), the Independencia Group of eastern Paraguay (Rapalini et al., 2006), and the Beaufort Group from South Africa (Lanci et al., 2013). The paleomagnetic pole from the Sierra Chica, La Pampa (Table 6) comes from a volcanic complex located in central Argentina (Colorado craton), which represents a smaller segment of the larger, Late Paleozoic –Early Mesozoic NW-SE trending volcano-plutonic belt of the Choiyoi Group (Domeier et al., 2011b; Llambías et al., 2003). Three U/Pb SHRIMP (zircon) ages of 263.0 ± 5.7 Ma, 268.1 ± 7.7 Ma, and 257.0 ± 2.8 Ma were obtained from the volcanic rocks, with an interpreted (median) age of 263 + 1.6/− 2.0 Ma (Domeier et al., 2011a). These ages, a mixture of intercept and Concordia age calculations, are not consistent with each other at the 95% uncertainty level, and cannot be considered a robust determination of the age of this unit beyond an assignment to the mid-Permian. The dual-polarity volcanic rocks pass both a reversal and fold test, and have been assigned a reliability factor of Q = 5 due

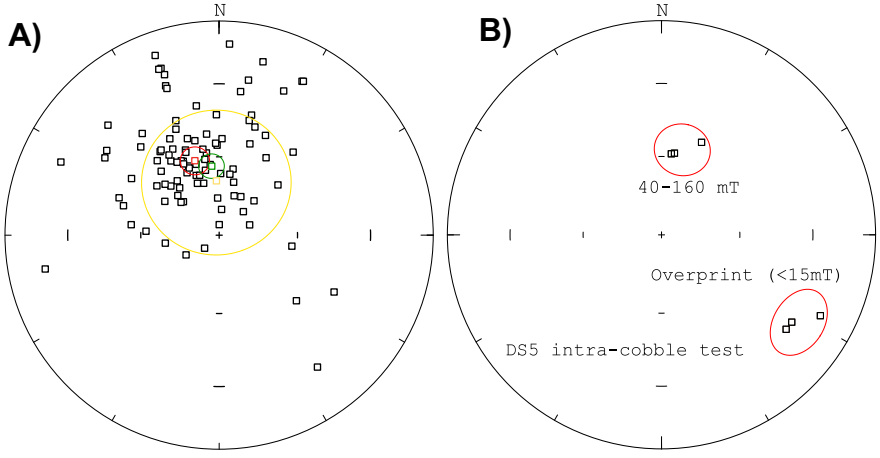


Fig. 9. (a) Stereoplot of the conglomerate test with the mean low temperature directions Components A (red) and DA (green) plotted for comparison. Up (down) pointing paleomagnetic directions are indicated by open (closed) squares in the stereoplots. The mean conglomerate direction (yellow) shows overlap in the α95 confidence cones of Components A and DA. (b) Intra-Cobble test from the volcanic dropstones, demonstrating remagnetization of the clasts subsequent to emplacement in the Kiama sandstone. Conglomerate test = negative, $R = 6.7$, $R0 = 4.7$, $R > R0$ non-random distribution (Watson, 1956). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

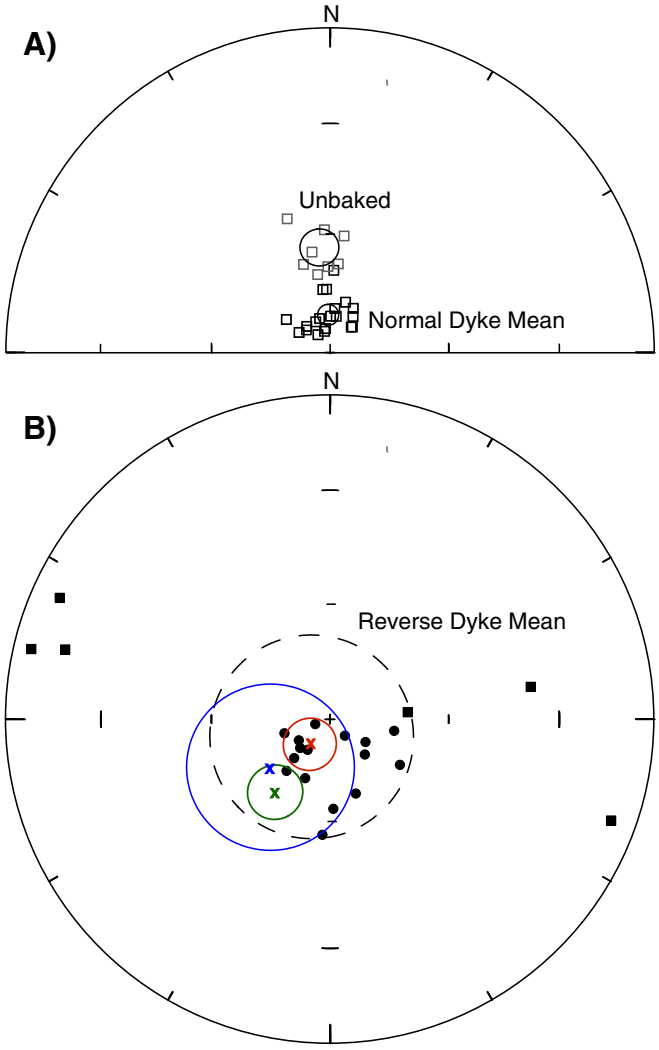


Fig. 10. Equal area diagrams displaying the a) mean normal polarity dyke direction and the mean unbaked direction, and b) the mean reverse dyke direction. Up (down) pointing paleomagnetic directions are indicated by open (closed) squares in the stereoplots. Arcconstraints for samples that do not reach the stable endpoints are calculated with the method of McFadden and McElhinny (1988) and are shown with circle symbols. Red: (Irving and Parry, 1963); Green: (Sakai et al., 2005); Blue: Component C of the Gerringong Volcanics. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to their similarity to Cretaceous paleomagnetic results (Domeier et al., 2011b; Somoza and Zaffarana, 2008). North of the La Pampa province lies the Mendoza province, which consists of ignimbrites, lavas, and subvolcanic porphyries of the Upper Choiyoi Group. An AMS and tilt-corrected dual-polarity paleomagnetic pole (Table 6) passes both reversal and fold tests, with a reliability factor of Q = 6. Two ages at 263.5 ± 2.0 Ma (SHRIMP, U/Pb, zircon) and 263.0 ± 2.0 Ma ($^{40}Ar/^{39}Ar$, biotite; recalculated from Domeier et al., 2011b; Renne et al., 2011) have been determined from the Upper Choiyoi group with an estimated age of ~264 Ma (Table 6; Domeier et al., 2011b).

Two reliable sedimentary poles which have been included in this compilation of Western Gondwana include the Independencia Group and the Beaufort Group. The Independencia Group consists of sub-horizontal sandstones and siltstones of the Parana Basin, South American craton (Rapalini et al., 2006). Paleomagnetic results are dual-polarity and pass a reversal test, and show no evidence for inclination shallowing from IRM experiments (Tan and Kodama, 2002; Tan et al., 2002). Full demagnetization of samples and good confidence of the mean (A_{95} = 6.6°) assigns the Independencia Group a reliability factor of Q = 5 (Van der Voo, 1990). The Independencia Group age assignment is based on the dual-polarity magnetization (post-KRS) and chronostratigraphic correlation of the Tacuray Formation (López-Gamundí et al., 1994), which overlies Late Carboniferous –Early Permian glacigenic deposits and underlies the Mesozoic Misiones Formation (Rapalini et al., 2006). The Beaufort Group of South Africa consists of Middle –Late Permian sandstones and siltstones of the Karoo Basin that have been corrected for inclination shallowing and pass a reversal test (Lanci et al., 2013; Tauxe and Kent, 2004; Tohver et al., 2015). The Beaufort Group sediments yielded four U/Pb (zircon) SHRIMP ages ranging from 264.6 ± 1.9 Ma to 268.5 ± 3.5 Ma (Lanci et al., 2013). Two normal magnetozones in the section have been correlated with the early Capitanian reversals of the GPTS, so an age of ca. 264 Ma is assigned to the paleomagnetic pole (Lanci et al., 2013; Ogg et al., 2016; Fig. 13). Due to the similarity of the normal polarity directions to the Jurassic paleomagnetic data the pole is assigned a reliability factor of Q = 5 (Ballard et al., 1986).

The results from Eastern Gondwana at ca. 265 Ma (Q = 6) overlap with the Middle Permian segment of the reference Gondwana APWP (Fig. 13), along with the well-constrained results from the Upper Choiyoi Group (Q = 6; ~264 Ma) and the Independencia Group (Q = 5; ~260 Ma) of the Colorado and South American cratons of Western Gondwana (Domeier et al., 2011c; Rapalini et al., 2006). The Sierra Chica results (Q = 6) from the Colorado craton have a low uncertainty about the mean (A_{95} = 3.3°) and are in agreement at the 95% confidence level with the Gerringong Volcanics (A_{95} = 9.1°). Fig. 13 illustrates this, and supports the interpretation of the SSB being representative of cratonic Australia within Gondwana.

The Beaufort Group pole (Q = 5) from South Africa (Fig. 13) is located further south than would be expected for a pole with an age of ~264 Ma (Lanci et al., 2013; Tohver et al., 2015). Since the Beaufort Group sediments have been corrected for inclination shallowing, it is likely that the observed discrepancy is related to inadequate removal of the normal polarity Jurassic overprint direction (Ballard et al., 1986). When only reverse (KRS) paleomagnetic data are considered, the lower Beaufort Group shows better agreement with the correct segment of the Gondwana APWP (37.8°S, 77.0°E, A_{95} = 11.7°, N = 22; calculated from Tohver et al., 2015).

5.3. Eastern Gondwana (270 –260 Ma) and the NEO

A comparison is made between the reliable Middle Permian paleomagnetic results of Western Gondwana (Q ≥ 5) and the available Eastern Gondwana paleomagnetic poles with an associated age in the range of 270 –260 Ma queried from the Global Paleomagnetic Database (Pisarevsky, 2005) in order to critically assess the Eastern Gondwanan data. Results with high error (A_{95} > 20°) or with an age range of more than ±20 Ma were not considered. Where multiple studies existed for the same rock unit, the most complete is used for comparison. Unfortunately there are no well-dated paleomagnetic results from India, Madagascar, and Timor with an age of 270–260 Ma, and the remainder of poles available are from eastern Australia. The analysis of the Australian data is not meant to classify the results as paleomagnetically viable for APWP considerations, but to illustrate potential issues with the available Middle Permian Eastern Gondwana data, propose solutions to the observed discrepancies, and compare with the new paleomagnetic result at ca. 265 Ma.

Uncertainty regarding whether Permian results from the NEO can be considered representative of cratonic Gondwana has led to their exclusion in the calculation of the Gondwana APWP (Domeier et al., 2012; McElhinny and McFadden, 2000; McElhinny et al., 2003; Torsvik et al., 2008; Torsvik et al., 2012). The NEO of eastern Australia has been affected by several complex deformation events, some of which resulted in the formation and progressive contortion of a complicated oroclinal structure (Klootwijk, 2003; Li and Rosenbaum, 2014; Musgrave, 2015; Shaanan et al., 2015a). The timing of these deformation events is mostly

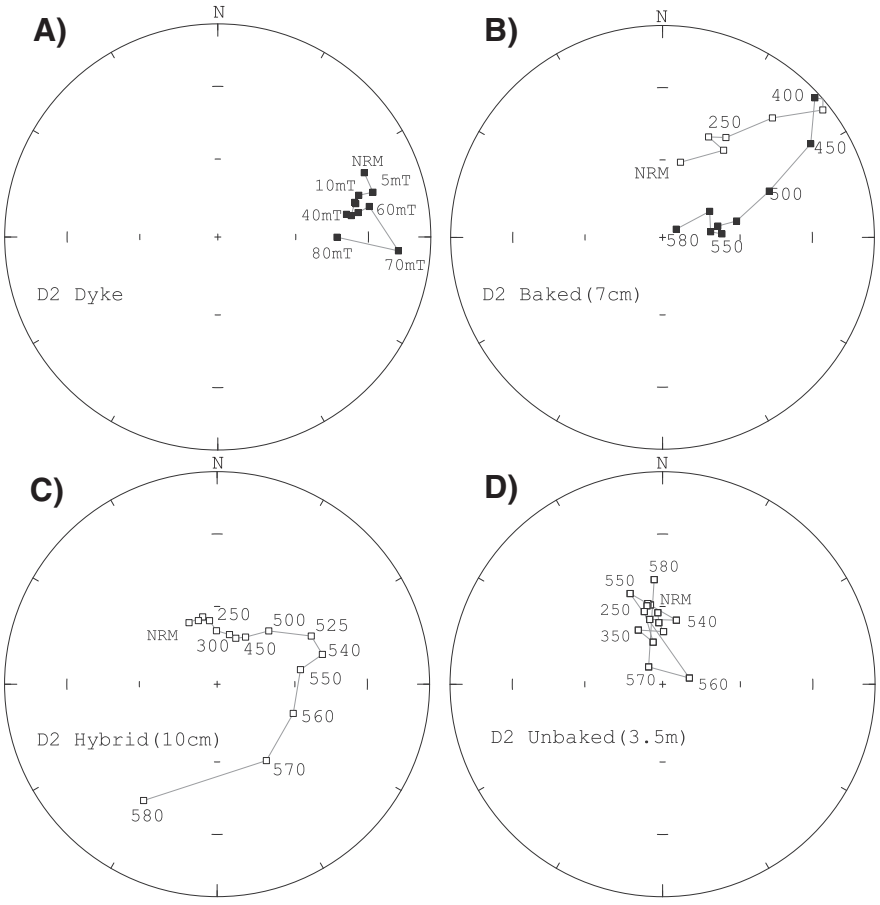


Fig. 11. Baked contact test from (a) Dyke 2 (reverse polarity) and host rock showing the (b) remagnetization of the Jamberoo sandstone 7 cm from the contact, (c) hybrid magnetization 10 cm from the contact, and (d) unbaked direction from the (distant) host rock. Solid (open) squares represent projections on the horizontal (vertical) plane in the orthogonal plots while up (down) pointing paleomagnetic directions are indicated by open (closed) squares in the stereoplots.

constrained through structural mapping, the emplacement and curvature of S-type granitoids, new U/Pb ages, paleomagnetic data, and theoretical plate motion models along the paleo-Pacific margin (Cawood et al., 2011; Klootwijk, 2003; Li and Rosenbaum, 2014; Li et al., 2014; Pisarevsky et al., 2016; Rosenbaum et al., 2012; 2014, Shaanan et al., 2015a, 2015b).

Before any paleomagnetic results can be included in the calculation of the Gondwana APWP, they must first be demonstrated to be representative of cratonic Australia. Pisarevsky et al. (2016) provide an updated list of Euler rotation parameters from 340 –270 Ma for the individual terranes of the NEO, which has been interpreted to have stabilized with cratonic Australia by 272 Ma (Shaanan et al., 2015a). However, further deformation of the NEO was recorded during the Permian –Triassic Hunter-Bowen orogeny as evidenced from structural mapping and paleomagnetic overprint directions of Carboniferous – Early Permian volcanic rocks (Holcombe et al., 1997; Klootwijk, 2003). It is, therefore, vital that any Middle Permian paleomagnetic data from the NEO be accompanied by the correct structural information and pass the appropriate field (fold) tests for primary magnetization.

Paleomagnetic reconnaissance work was conducted by Klootwijk (2003) from nine sites of the Werrie Basalt and six sites of the Boggabri Volcanics in the north-western Tamworth Terrane (Table 7; Fig. 13). The Boggabri Volcanics are located in the Gunnedah Bain, west of the Mooki Thrust, while the Werrie Basalt is located about 100 km east in

Table 5
AMS results.

Site	N	K_m	L	F	P	T	U	K_{max}D	K_{max}I	K_{int}D	K_{int}I	K_{min}D	K_{min}I
B	5	0.0780	1.008	1.016	1.025	0.352	0.347	180.3	10.8	73.0	57.4	276.6	30.4
B1	8	0.0872	1.014	1.021	1.035	0.207	0.199	150.1	29.0	52.9	12.6	302.0	57.9
B3	7	0.0793	1.015	1.018	1.034	0.108	0.100	116.5	9.8	211.5	26.7	8.1	61.3
B4	7	0.0803	1.012	1.022	1.034	0.293	0.287	117.9	20.1	215.1	18.9	344.9	61.8
B5	6	0.0806	1.017	1.031	1.049	0.298	0.287	105.9	11.6	330.0	74.0	198.1	10.8
Bumbo Combined	**33**	**0.0814**	**1.013**	**1.022**	**1.036**	**0.243**	**0.235**	**121.9**	**22.1**	**216.9**	**12.0**	**333.3**	**64.6**
B7	6	0.0521	1.019	1.012	1.032	−0.237	−0.244	261.4	13.5	168.5	11.9	38.4	71.8
B8	7	0.0531	1.014	1.019	1.033	0.121	0.116	159.2	29.3	263.3	23.5	25.7	50.9
B9	8	0.0630	1.011	1.011	1.022	−0.048	−0.053	225.3	17.1	122.0	36.7	335.5	48.2
B11	10	0.0655	1.014	1.009	1.023	−0.249	−0.254	192.8	25.3	288.8	12.5	42.9	61.4
Dapto Combined	**32**	**0.0595**	**1.014**	**1.012**	**1.027**	**−0.111**	**−0.117**	**193.7**	**26.8**	**284.4**	**1.4**	**17.2**	**63.2**

Notes: N = number of samples; K_m = average bulk volume susceptibility in SI units; L = Magnetic lineation: K_{max}/K_{int}; F = foliation: K_{int}/K_{min}; P = anisotropy degree: K_{max}/K_{int}; T = Shape factor: $T = 2ln(K_{int}/K_{min})/ln(K_{max}/K_{min}) - 1$; U = Difference shape factor: $(2K_{int}\text{-}K_{max}\text{-}K_{int})/(K_{max}\text{-}K_{int})$; K_{max} (K_{min}) Dec/Inc. = Declination/Inclination. Mean tensor directions for the Bumbo and Dapto Latites were calculated from combined sample-level data.

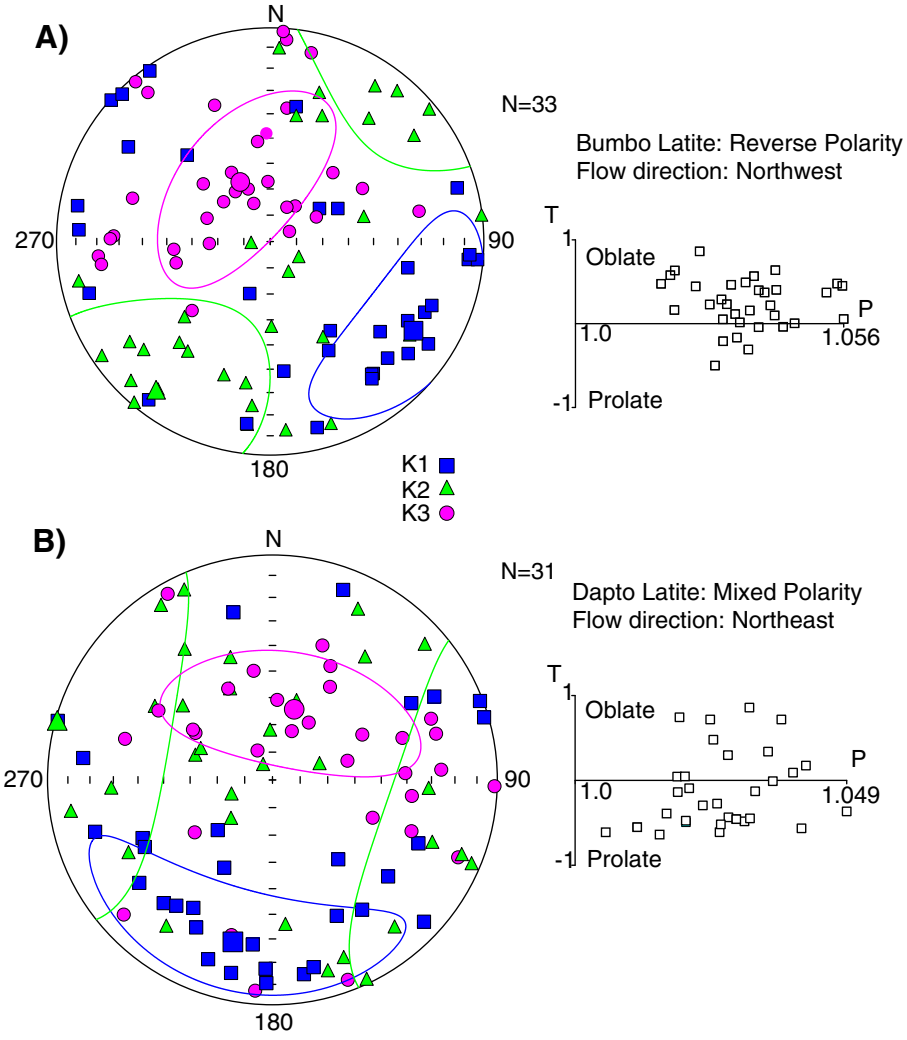


Fig. 12. Principal axes of magnetic susceptibility for the (a) Bumbo and (b) Dapto Latites. Left: Equal-area plots of the directions of the principal axes of AMS. Right: Jelinek diagrams displaying the general shape of the AMS ellipsoids with P = degree of anisotropy and T = shape factor. K1 = Kmax, K2 = Kint, K3 = Kmin. Ovals represent the confidence ellipses for the mean direction.

the southern NEO. An age of 266.4 ± 3.0 Ma ($^{40}Ar/^{39}Ar$, whole rock, isochron) has been reported from the pyroclastic rocks and ignimbrites of the Werrie Basalt, as well as a Rb/Sr age of 269.4 ± 4.6 Ma (Flood et al., 1988) from the Warrigundi Igneous Complex (interpreted volcanic centre). The flow-banded lavas and lithic tuffs of the Boggabri Volcanics are assigned an age of 260 –299 Ma as listed in the Global Paleomagnetic Database. This age assignment comes from palynological data and a presumed primary KRS magnetization (Klootwijk, 2003). Fig. 13 demonstrates good agreement between the Boggabri Volcanics and the reference Gondwana APWP, as well as the Western Gondwanan results and the Gerrigong Volcanics. Based on the available reliable paleomagnetic data it is likely that the Boggabri Volcanics are representative of cratonic Australia and Gondwana at 265 Ma. A reliability factor of Q = 5 is assigned to the Boggabri Volcanics because of a lack of reversals as well as a precise age. The Werrie Basalt on the other hand, has a much larger error (A_{95} = 16°), which may be related to limited structural information or uncertainty regarding the averaging of secular variation for the individual cooling unit (Klootwijk, 2003; Lackie and Schmidt, 1993), so a reliability factor of Q = 5 is applied (Van der Voo, 1990). The Werrie Basalt pole plots about 15° north of the Boggabri Volcanics, with a critical angle of 21.4° (angle between the mean directions of the two sets of observations at which the null hypothesis of a common mean direction would be rejected with 95% confidence) between the raw 270 Ma mean pole of the Gondwana APWP (McFadden and McElhinny, 1990; Torsvik et al., 2012). The difference in directions may be due to several possibilities. One possibility is that the Werrie Basalt is older than the Boggabri Volcanics, and that the reported ages might be too young; another possibility is that tectonic rotation in the NEO extended to the Middle Permian, and that this deformation was restricted to east of the Mooki Thrust.

A recent tectonic study by Shaanan et al. (2015a) explored the paleomagnetism of the Alum Mountain Volcanics of the Gloucester Syncline (Myall Block). This paleomagnetic pole from the NEO has an assigned age of ca. 272 Ma, and pre-dates compression associated with the Hunter-Bowen Orogeny (Holcombe et al., 1997; Li et al., 2014; Roberts et al., 1996). The Alum Mountain Volcanics consist of basaltic to rhyolitic flow deposits and ash-fall tuffs, and yielded two early- to mid-Permian ages of 271.8 ± 1.8 Ma ($^{40}Ar/^{39}Ar$, whole rock, inverse isochron; Li et al., 2014) and 274.1 ± 3.4 Ma (U/Pb, SHRIMP, zircon; Roberts et al., 1996, reliant upon the heterogeneous SL13 zircon standard). The Alum Mountain Volcanics pass a fold test, however, the high uncertainty associated with the paleomagnetic pole (A_{95} = 19.5°), as well as the well-documented episodes of post-Early Permian structural translation in the Tamworth Belt of the NEO, assigns the pole a reliability factor of Q = 3 (Shaanan et al., 2015a). A critical angle of 29.6° between the Alum Volcanics pole and the 270 Ma mean pole of the Gondwana APWP suggests that the rocks were subjected to tectonic rotation relative to cratonic Gondwana (Fig. 13). Interestingly, the paleomagnetic pole was proposed to pre-date the Hunter-Bowen

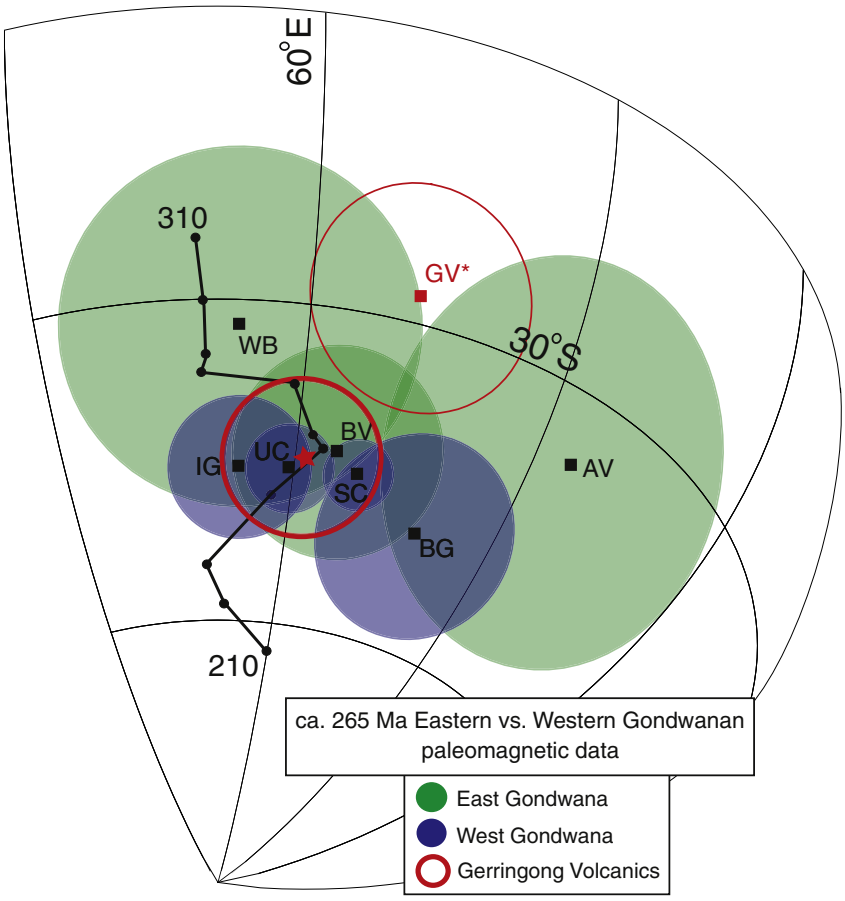


Fig. 13. A paleomagnetic comparison of available ca. 265 Ma poles and associated A95 confidence circles from Western Gondwana (blue), Eastern Gondwana (green), and the Gerringong Volcanics (red). The recommended APWP for Gondwana is plotted only for reference (Torsvik et al., 2012). The Gerringong Volcanics overlap with recent results from the Upper Choiyoi Group of South America (Western Gondwana; Domeier et al., 2011b). All poles were rotated into South African coordinates using the recommended Euler parameters (Torsvik et al., 2008, 2012). Star = Gerringong Volcanics (this study); GV* = Gerringong Volcanics (Irving and Parry, 1963); WB = Werrie Basalt; RV = Rookwood Volcanics; AV: Alum Mtn. Volcanics; BV = Boggabri Volcanics; UC = Upper Choiyoi Group; IG = Independencia Group; SC = Sierra Chica, La Pampa; BG = Beaufort Group (See Tables 6 and 7 for a list of pole references). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

orogeny and post-date the final stages of oroclinal bending (270 Ma). The discrepancies between the paleomagnetic and geologic data may be a result of imprecise age dating, though it is noted that disagreement remains on the timing of deformation in the NEO (Musgrave, 2015).

The remaining result is from the original study of the Gerringong Volcanics (Irving and Parry, 1963), which lies outside of the A_{95} confidence cones of the Western Gondwana poles and the present study. As noted in Section 4.2, this is due to the unresolved secondary

Table 6
Paleomagnetic poles for a ca. 265 Ma West Gondwana (Africa, South America).

Formation	Polarity	Craton	Age (Ma)	Method	Lat	Lon	R_{lat}	R_{lon}	Euler	A_{95}	Q	Ref
Sierra Chica, La Pampa	Mixed	Colorado	~263	263.0 + 1.6/−2.0 Ma (SHRIMP, U/Pb, zircon; median age)	−80.1	349.0	−44.8	68.6	(47.5, −33.3, 57.3)	3.3	0111110	1
Upper Choiyoi Group	Mixed	Colorado	~264	263.5 ± 2.0 Ma (SHRIMP, U/Pb, zircon); 263.0 ± 2.0 Ma ($^{40}Ar/^{39}Ar$, biotite; recalculated from Domeier et al., 2011b; Renne et al., 2011)	−73.7	315.6	−46.6	55.0	(47.5, −33.3, 57.3)	4.1	1111110	2
Beaufort Group	Mixed	S. Africa	264	264.6 ± 1.9 Ma (SHRIMP, U/Pb, zircon); magnetostratigraphic correlation	−48.9	78.8	−48.9	78.8		9.7	1110110	3, 4
Independencia Group	Mixed	S. America	~260	Post-KRS; chronostratigraphic correlation	−80.7	7.0	−45.5	52.9	(50,/−32.5/55.1)	6.6	0111110	5

Notes: Lat = Pole Latitude, Lon = Pole Longitude, A_{95} = radius of the 95% confidence circle about the calculated mean pole, Q = Quality Factor (Van der Voo, 1990), R_{lat}/R_{lon} = pole rotated to South African coordinates using the recommended Euler plate rotations (Torsvik et al., 2008, 2012). Ref: 1 = Domeier et al. (2011a); 2 = Domeier et al. (2011b); 3 = Lanci et al. (2013); 4 = Tohver et al. (2015); 5 = Rapalini et al. (2006).

Table 7
Paleomagnetic poles for a ca. 265 Ma East Gondwana (Australia).

Rock name (ChRM)	Polarity	Region	Age (Ma)	Source	Lat	Long	R_{lat}	R_{long}	A_{95}	Q	Ref
Werrie Basalt	Reverse	NEO	~266	266.4 ± 3.0 Ma ($^{40}Ar/^{39}Ar$; whole rock; isochron)	−61.2	128.7	−32.4	52.5	16.0	1,111,001	1, 2
Boggabri Volcanics	Reverse	NEO	260 − 299	Palynology	−53.6	152.6	−43.2	65.4	10.0	0111101	1
Alum Mtn. Volcanics	Reverse	NEO	~272	271.8 ± 1.8 Ma ($^{40}Ar/^{39}Ar$, whole rock, inverse isochron); 274.1 ± 3.4 (U/Pb, SHRIMP, zircon)	−30.0	153.2	−36.7	95.2	19.5	0011001	1, 4, 5
Gerringong Volcanics	**Mixed**	**SSB**	**265**	**265.05 ± 0.35 [0.46] ($^{40}Ar/^{39}Ar$, plagioclase; Bumbo); 263.51 ± 0.05 [0.31] Ma (U/Pb TIMS, zircon, Broughton tuff)**	**−56.9**	**154.8**	**−44.8**	**61.0**	**9.1**	**1,111,110**	**6, 7**
	Reverse			*Previous study*	*−44.0*	*132.0*	*−27.5*	*71.7*	*11.0*	0100101	*8*

Notes: Lat = Pole Latitude, Lon = Pole Longitude, A_{95} = radius of the 95% confidence circle about the calculated mean pole, Q = Quality Factor (Van der Voo, 1990), R_{lat}/R_{lon} = pole rotated to South African coordinates using the recommended Euler plate rotation: 19.5, 117.8, −56.2 (Torsvik et al., 2008). NEO = New England Orogen; SSB = southern Sydney Basin. Ref: 1 = Klootwijk (2003); 2 = Li et al. (2014); 3 = Clark (1994); 4 = Shaanan et al. (2015b); 5 = Roberts et al. (1996); 6 = This study; 7 = Metcalfe et al. (2015); 8 = Irving and Parry (1963).

components of magnetization common in the partial demagnetization techniques of older paleomagnetic studies. However, if modern demagnetization methods are applied with good resolution of individual paleomagnetic components, and the correct structural and age information can be obtained (e.g. the Boggabri Volcanics), then the dataset of East Gondwana can be reconciled with that of Western Gondwana.

5.4. Paleomagnetic test for a Middle Permian (270–260 Ma) Pangea

Since the original proposal for Pangea B, much work has been conducted in exploring non-dipole fields, inclination shallowing, and providing new isotopic ages with paleomagnetic data (Domeier et al., 2011b, 2011c, 2012; Irving, 1977; Muttoni et al., 2003; Torsvik and Van der Voo, 2002; Torsvik et al., 2012; Van der Voo and Torsvik, 2001). A Pangea B configuration was originally proposed to pre-date the Late Triassic, but this estimate has been revised with the continued publication of high quality Permian-Triassic paleomagnetic data (Domeier et al., 2011b, 2011c, 2012; Dominguez et al., 2011; Muttoni et al., 2009). Disagreement remains on the Middle Permian paleogeography of Pangea, with recent work advocating an Early Permian Pangea B (296–272 Ma), with a transition from 272–260 Ma to a Permian-Triassic Pangea A (260–249 Ma) along a 3000 km intra-Pangean dextral shear (Fig. 3; Muttoni et al., 2009). Some geologic evidence has been provided that may support this interpretation (Aubele et al., 2012; Muttoni et al., 2009, 2003), however the availability of well-dated Middle Permian paleomagnetic results has been very limited prior to the last 5 years.

The Middle Permian Pangea B configuration proposed by Muttoni et al. (2009) is an interpolation between the previously available Early Permian and Permian-Triassic paleomagnetic data, with relative paleolatitudes of the continents constrained by the Gerringong Volcanics (Irving and Parry, 1963), the Kama region sediments (Khramov, 1982), and the Thini Chu Group of Nepal (Klootwijk and Bingham, 1980). There are a few problems with these constraints. The original study of the Gerringong Volcanics only applied partial demagnetization, and due to unresolved overprint components is displaced by about 15° north of the result presented here (Irving and Parry, 1963). The Thini Chu Group pole of Nepal is based on only one site, and there are no good age constraints other than stratigraphic considerations (Klootwijk and Bingham, 1980). The pole is also restricted to sedimentary rocks, which should be avoided unless a test can be conducted for potential inclination shallowing. The Kama region pole from the Russian platform of Europe was used as a paleolatitudinal constraint for Northern Pangea, and was also restricted to sedimentary rocks with a broad age assignment of 271–251 Ma based on fossil assemblages (Khramov and Slautsitais, 1982). Fortunately, several well-dated Middle Permian paleomagnetic results have been published in the last few years from Laurentia, Baltica, and Western Gondwana, and can be combined with the Eastern Gondwana result presented here to provide a paleomagnetic test of Pangea (Domeier et al., 2011a, 2011c; Dominguez et al., 2011).

Reconstruction poles for a ca. 265 Ma Pangea were restricted to igneous results with Q ≥ 5 and a well determined radio-isotopic age of 265 ± 5 Ma (Table 8). These include the Upper Choiyoi Group of South America (West Gondwana), the Illinois intrusions of North America (Laurentia), the Lunner Dykes (Stable Europe), and the Gerringong Volcanics (East Gondwana). Since there are no well-dated igneous results from this interval in Antarctica, Africa, India, and

Table 8
Reconstruction poles for a Middle Permian (ca. 265 Ma) test of Pangea.

Formation	Continent	Age (Ma)	Method	Lat	Lon	A_{95}	Q	Ref
Upper Choiyoi Group	West Gondwana	~264	263.5 ± 2.0 Ma (SHRIMP, U/Pb, zircon); 263.0 ± 2.0 Ma ($^{40}Ar/^{39}Ar$, biotite; recalculated from Domeier et al., 2011b; Renne et al., 2011)	−73.7	315.6	4.1	6	1
Gerringong Volcanics	East Gondwana	265	265.05 ± 0.35 [0.46] ($^{40}Ar/^{39}Ar$, plagioclase; Bumbo); 263.51 ± 0.05 [0.31] Ma (U/Pb TIMS, zircon, Broughton tuff)	−56.9	154.8	9.1	6	2
Illinois Intrusions	Laurentia	~270	267.8–273.6 Ma = Range of published mean age interpretations (calculated in Domeier et al., 2011c; see references therein)	−56.3	302.9	3.8	5	3
Lunner dykes	Stable Europe	~270	Combined $^{40}Ar/^{39}Ar$ weighted mean age of 271 ± 2.7 Ma (Calculated in Dominguez et al., 2011; see references therein)	−50.9	343.8	3.5	5	4

Notes: Lat = Pole Latitude, Lon = Pole Longitude, A_{95} = radius of the 95% confidence circle about the calculated mean pole, Q = Quality Factor (Van der Voo, 1990). Ref: 1 = Domeier et al. (2011b); 2 = This Study; 3 = Domeier et al. (2011c) 4 = Dominguez et al. (2011).

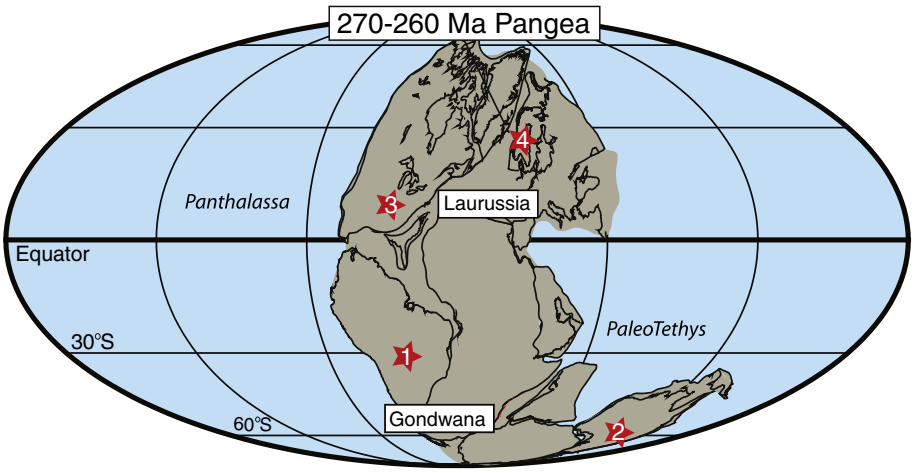


Fig. 14. Paleomagnetic reconstruction of a Middle Permian Pangea (270–260 Ma) using the poles (1 –4) listed in Table 8. Continents were plotted using a Mollweide projection in GMAP 2003 (TH Torsvik). The paleolatitude of Gondwana relative to Laurussia supports the Pangea A configuration for this interval, with a relatively loose fit between the respective continental margins.

Madagascar, the Eastern Gondwanan continents were fixed relative to Australia, while the Western Gondwanan continents were fixed relative to South America using the recommended Euler rotation poles of Torsvik et al. (2012). The resulting paleogeographic reconstruction is consistent with the Wegenerian Pangea configuration (Fig. 14), with the north-western coast of Africa against the south-eastern coast of Laurentia (Bullard et al., 1965). The relative paleolatitudes of the continents show a loose N-S fit between Laurussia and Gondwana, with no evidence for crustal overlap, and confirms recent work from Western Gondwana, Laurentia, and Baltica in support of a Middle Permian Pangea A (Domeier et al., 2011a, 2011c; Dominguez et al., 2011). We conclude that if Pangea B existed prior to Pangea A, than it must have finished assembly by ca. 265 Ma.

5.5. The Illawarra Reversal

There is abundant magnetostratigraphic evidence that, from ca. 318 to 265 Ma, Earth's magnetic field remained fixed in a reverse polarity (Alva-Valdivia et al., 2002; Diehl and Shive, 1981, 1979; DiVenere and Opdyke, 1991, 1990; Gialanella et al., 1997; Gose and Helsley, 1972; Irving and Parry, 1963; Khramov and Tarling, 1987; Lozovsky and Molostovsky, 1993; McFadden et al., 1988; Menning et al., 1988; Nawrocki, 1997; Opdyke and Channell, 1996; Opdyke et al., 2001, 2000; Steiner, 2006; Steiner et al., 1993; Szurlies et al., 2003). The base of the KRS (Patterson Reversal) was well documented by Opdyke et al. (2000) from volcanic and sedimentary rocks of the Tamworth Belt; however, the top of the superchron (Illawarra Reversal) has not been directly dated. Although commonly stated to be ~265 Ma in numerous publications, no published data exist that correlate a precise radio-isotopic age with the appropriate polarity transition. Steiner (2006) provided a review of Permian KRS magnetostratigraphic data, including short normal polarities observed in the Goose Egg Formation in Wyoming (Renner, 1988) and unpublished data from the Queen Formation of Texas and New Mexico. Poor age resolution, discontinuity of sampled exposures, and uncertainty regarding overprint magnetizations, means only an approximate age of 267 Ma was proposed for the Illawarra Reversal (Fig. 8; Steiner, 2006).

Lanci et al. (2013) recently described a Middle Permian magnetostratigraphic section containing normal polarities from the Beaufort Group, in which the normal magnetozones were interpreted to coincide with the Capitanian reversals of the GPTS (Ogg et al., 2016) and an age assignment of 269 Ma was proposed for the end of the KRS (Fig. 8). Middle Permian normal polarities have also been detected in SW England and Japan (Hounslow et al., 2016; Kirschvink et al., 2015) and have been assigned a Wordian age following the compilation of Steiner (2006). The biostratigraphic zones in Japan are defined by Tethyan fusulinid assemblages (Kasuya et al., 2012), which have been tentatively correlated with the North American conodont-defined International Stratigraphic Stages of the Guadalupian (Wilde, 1999). There remains an outstanding issue regarding the reliability of biostratigraphic correlation between the Guadalupian North American and Tethyan domains (Leven, 2001), so it is likely that this interval is slightly younger.

The Rotliegend Group of the Central European Basin records the largest sampled stratigraphic interval through the KRS (Menning et al., 1988). This section shows an absence of normal polarities prior to the late Capitanian (Menning et al., 1988). Further work by Menning (2000) and Menning et al. (2006) in the Guadalupian type section of western Texas suggested a ~265 Ma age for the end of the KRS. The corresponding magnetostratigraphic data for this section are unpublished, however, and the area has been reported as unsuitable for paleomagnetism (Steiner, 2006).

This study of the Gerringong Volcanics provides a normal polarity reversal with a corresponding radio-isotopic age from the KRS type section in eastern Australia. The age recorded in the (Normal) Saddleback and Dapto Latites stratigraphically overlie the (Reverse) Bumbo (265.05 ± 0.35 [0.46]) and Blow Hole Latites. Furthermore, the recent addition of dual-polarity Western Gondwanan poles at ca. 263 Ma (Domeier et al., 2011b) and ca. 264 Ma (Domeier et al., 2011c), as well as the reverse polarity Eastern Gondwanan poles at ca. 266 Ma (Klootwijk, 2003; Li et al., 2014) and ca. 272 Ma (Li et al., 2014; Roberts et al., 1996; Shaanan et al., 2015a) are consistent with this age assignment.

An analysis of the Global Paleomagnetic Database from 318 to 265 Ma also supports an absence of normal magnetozones older than 265 Ma, however several short and undated normal polarity sub-chrons have been reported within the KRS (Diehl and Shive, 1981, 1979; Gose and Helsley, 1972; Halverson et al., 1989; Helsley, 1965; Hounslow and Balabanov, 2016; Irving and Monger, 1987; Menning et al., 1988; Miller and Opdyke, 1985; Sinito et al., 1979; Symons, 1990; Wynne et al., 1983). A 700 m thick, continuous section from the Patlanoaya Formation in Mexico supports previous estimates of Early Permian (~285 Ma) and upper Pennsylvanian (~300 Ma) normal sub-chrons (Alva-Valdivia et al., 2002; Menning et al., 1988; Ogg et al., 2016; Sinito et al., 1979). Reverse sub-chrons have also been recorded in the ~37 Ma Cretaceous Normal Superchron (Lowrie et al., 1980; Tarduno, 1990; VandenBerg et al., 1978). Although normal polarities may be detected within the KRS, the magnetostratigraphy from the

Central European Basin confirms the primarily reverse polarity of Earth's magnetic field for the Permo-Carboniferous (Menning et al., 1988). The presence of superchrons during the Phanerozoic indicates that the geomagnetic field likely has two states, one with reversal frequencies exceeding one per million years and the other with no or very few reversals for tens of millions of years (McElhinny and McFadden, 2000).

It is likely that the Wordian-Capitanian boundary (ca. 265 Ma) marks the transition from a non-reversing dynamo to a reversing one, with longer intervals of stable normal polarity (magnetozones) recorded and correlated globally (Opdyke and Channell, 1996; Steiner, 2006). Short and sporadic normal polarities reported within the KRS may represent typical non-reversing geomagnetic field behaviour, however, the geodynamics of field reversals are still poorly understood (Granot et al., 2012; Hounslow and Balabanov, 2016; Lhuillier et al., 2016). The addition of new Permian and Carboniferous paleomagnetic data will help resolve the geomagnetic behaviour of superchrons such as the KRS.

6. Conclusions

This study aimed to provide better paleomagnetic resolution of the Middle Permian with respect to paleogeographic and global magnetostratigraphic data. Determination of a paleomagnetic pole from Eastern Gondwana has allowed us to make a comparison with the Global Paleomagnetic Database and the Global Magnetic Polarity Timescale to improve our understanding of the complex tectonic and geodynamic events for this period.

1. Paleomagnetic data from volcanic rocks of the southern Sydney Basin of Eastern Gondwana show statistical overlap with the reference Gondwana APWP of Torsvik et al. (2012) and recent reliable paleomagnetic poles from Western Gondwana at ca. 265 Ma. The resolution of a dual-polarity ChRM direction (Components B_N and B_C) free from secondary magnetizations demonstrates that the observed longitudinal discrepancies between the Western and Eastern Gondwanan datasets can be reconciled through the re-examination of older paleomagnetic results.
2. A selection of well-dated Middle Permian paleomagnetic results from Western Gondwana, Laurentia, Eastern Gondwana, and Baltica support the classical Wegenerian Pangea A configuration, with a loose N-S fit of Gondwana and Laurussia. If a transition from Pangea B to A occurred along the proposed dextral shear zone, then it must have been completed prior to the Middle Permian (ca. 265 Ma).
3. A normal polarity reversal post-dating the KRS, with a $^{40}Ar/^{39}Ar$ plagioclase age of 265.05 ± 0.35 [0.46] Ma, supports an age of 265 Ma for the end of the KRS. The Illawarra Reversal provides a prominent magnetostratigraphic marker that may be linked with the biostratigraphically constrained geologic timescale to provide a reliable reference for global correlation of Paleozoic sedimentary sections.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.tecto.2016.12.029.

References

Alva-Valdivia, L.M., Goguitchaichvili, A., Grajales, M., de Dios, A.F., Urrutia-Fucugauchi, J., Rosales, C., Morales, J., 2002. Further constraints for Permo-Carboniferous magnetostratigraphy: case study of the sedimentary sequence from San Salvador–Patlanoaya (Mexico). Compt. Rendus Geosci. 334 (11), 811–817.
Anson, G., Kodama, K., 1987. Compaction-induced inclination shallowing of the post-depositional remanent magnetization in a synthetic sediment. Geophys. J. Int. 88 (3), 673–692.
Aubele, K., Bachtadse, V., Muttoni, G., Ronchi, A., Durand, M., 2012. A paleomagnetic study of Permian and Triassic rocks from the Toulon-Cuers Basin, SE France: evidence for intra-Pangea block rotations in the Permian. Tectonics 31 (3).
Bachtadse, V., Zänglein, R., Tait, J., Soffel, H.C., 2002. Palaeomagnetism of the Permo/Carboniferous (280 Ma) Jebel Nehoud ring complex, Kordofan, Central Sudan. J. Afr. Earth Sci. 35 (1), 89–97.
Ballard, M.M., Van der Voo, R., Hälbich, I., 1986. Remagnetizations in Late Permian and Early Triassic rocks from southern Africa and their implications for Pangea reconstructions. Earth Planet. Sci. Lett. 79 (3-4), 412–418.
Black, L.P., Kamo, S.L., Williams, I.S., Mundil, R., Davis, D.W., Korsch, R.J., Foudoulis, C., 2003. The application of SHRIMP to Phanerozoic geochronology; a critical appraisal of four zircon standards. Chem. Geol. 200 (1), 171–188.
Bowman, H., 1974. Geology of the Wollongong, Kiama, and Robertson 1: 50,000 Sheets: Sheets 9029-II, 9028-I and IV. Geological Survey of New South Wales.
Bowring, S.A., Erwin, D., Jin, Y., Martin, M., Davidek, K., Wang, W., 1998. U/Pb zircon geochronology and tempo of the end-Permian mass extinction. Science 280 (5366), 1039–1045.
Briden, J., Smith, A.G., Sallomy, J., 1971. The geomagnetic field in Permo-Triassic time. Geophys. J. Int. 23 (1), 101–117.
Briden, J., Hurley, A., Smith, A., 1981. Paleomagnetism and Mesozoic-Cenozoic paleocontinental maps. J. Geophys. Res. Solid Earth 86 (B12), 11631–11656.
Bullard, E., Everett, J.E., Smith, A.G., 1965. The fit of the continents around the Atlantic. Philos. Trans. R. Soc. Lond. A: Math. Phys. Eng. Sci. 258 (1088), 41–51.
Butler, R.F., 1992. Paleomagnetism: Magnetic Domains to Geologic Terranes. Blackwell Scientific Publications.
Campbell, L., Conaghan, P., Flood, R., 2001. Flow-field and palaeogeographic reconstruction of volcanic activity in the Permian Gerringong Volcanic Complex, southern Sydney Basin, Australia. Aust. J. Earth Sci. 48 (3), 357–375.
Carr, P., 1983. A reappraisal of the stratigraphy of the upper Shoalhaven group and lower Illawarra coal measures, southern Sydney Basin, New South Wales. Proc. Linnean Soc. NSW 106, 287–297.
Carr, P., 1985. Geochemistry of Late Permian shoshonitic lavas from the southern Sydney Basin. In: Sutherland, F.L., Franklin, B.J., Waltho, A.E. (Eds.), Proceedings, Volcanism in Eastern Australia with Case Histories from New South Wales, Geological Society of Australia, New South Wales Division, Sydney, pp. 165–183.
Carr, P., 1998. Subduction-related late Permian shoshonites of the Sydney basin, Australia. Mineral. Petrol. 63 (1-2), 49–71.
Carr, P., Pemberton, J., Nunan, E., 1999. Low-grade metamorphism of mafic lavas, Upper Permian Broughton Formation, Sydney Basin. Aust. J. Earth Sci. 46 (6), 839–849.
Cawood, P., Pisarevsky, S., Leitch, E., 2011. Unraveling the New England orocline, east Gondwana accretionary margin. Tectonics 30 (5).
Chadima, M., Hrouda, F., 2009. Cureval 8.0: Thermomagnetic Curve Browser for Windows. Agico, Inc.
Clark, D., 1994. Magnetic Petrophysics and Palaeomagnetism of the Fitzroy Leases, Central Queensland—Implications for Exploration. CSIRO Australian Exploration and Mining Report, 6, p. 28.
Cottrell, R.D., Tarduno, J.A., Roberts, J., 2008. The Kiaman reversed polarity superchron at Kiama: toward a field strength estimate based on single silicate crystals. Phys. Earth Planet. Inter. 169 (1), 49–58.
Diehl, J.F., Shive, P.N., 1979. Palaeomagnetic studies of the Early Permian Ingelside formation of northern Colorado. Geophys. J. Int. 56 (2), 271–282.
Diehl, J.F., Shive, P.N., 1981. Paleomagnetic results from the Late Carboniferous/Early Permian Casper formation: implications for northern Appalachian tectonics. Earth Planet. Sci. Lett. 54 (2), 281–292.
DiVenere, V., Opdyke, N., 1990. Paleomagnetism of the Maringouin and Shepody formations, New Brunswick: a Namurian magnetic stratigraphy. Can. J. Earth Sci. 27 (6), 803–810.
DiVenere, V., Opdyke, N., 1991. Magnetic polarity stratigraphy in the uppermost Mississippian Mauch Chunk formation, Pottsville, Pennsylvania. Geology 19 (2), 127–130.
Domeier, M., Van der Voo, R., Denny, F.B., 2011a. Widespread inclination shallowing in Permian and Triassic paleomagnetic data from Laurentia: support from new paleomagnetic data from Middle Permian shallow intrusions in southern Illinois (USA) and virtual geomagnetic pole distributions. Tectonophysics 511 (1), 38–52.
Domeier, M., Van der Voo, R., Tohver, E., Tomezzoli, R.N., Vizan, H., Torsvik, T.H., Kirshner, J., 2011b. New Late Permian paleomagnetic data from Argentina: refinement of the apparent polar wander path of Gondwana. Geochem. Geophys. Geosyst. 12 (7).
Domeier, M., Van der Voo, R., Tomezzoli, R.N., Tohver, E., Hendriks, B.W., Torsvik, T.H., Vizan, H., Dominguez, A., 2011c. Support for an "A-type" Pangea reconstruction from high-fidelity Late Permian and Early to Middle Triassic paleomagnetic data from Argentina. J. Geophys. Res. Solid Earth 116 (B12).
Domeier, M., Van der Voo, R., Torsvik, T.H., 2012. Paleomagnetism and Pangea: the road to reconciliation. Tectonophysics 514, 14–43.
Dominguez, A.R., Van der Voo, R., Torsvik, T.H., Hendriks, B.W., Abrajevitch, A., Domeier, M., Larsen, B.T., Rousse, S., 2011. The ~270 Ma palaeolatitude of Baltica and its significance for Pangea models. Geophys. J. Int. 186 (2), 529–550.

Dunlop, D., Özdemir, Ö., Schmidt, P., 1997a. Paleomagnetism and paleothermometry of the Sydney Basin 2. Origin of anomalously high unblocking temperatures. J. Geophys. Res. Solid Earth 102 (B12), 27285–27295.
Dunlop, D., Schmidt, P., Özdemir, Ö., Clark, D., 1997b. Paleomagnetism and paleothermometry of the Sydney Basin 1. Thermoviscous and chemical overprinting of the Milton Monzonite. J. Geophys. Res. Solid Earth 102 (B12), 27271–27283.
Everitt, C., Clegg, J., 1962. A field test of palaeomagnetic stability. Geophys. J. Int. 6 (3), 312–319.
Evernden, J., Richards, J., 1962. Potassium-argon ages in eastern Australia. J. Geol. Soc. Aust. 9 (1), 1–49.
Facer, R., Carr, P., 1979. K-Ar dating of Permian and Tertiary igneous activity in the south-eastern Sydney Basin, New South Wales. J. Geol. Soc. Aust. 26 (1-2), 73–79.
Fisher, R., 1953. Dispersion on a sphere. Proc. R. Soc. Lond. A Math. Phys. Sci. 217 (1130), 295–305.
Flood, R., Craven, S., Elmes, D., Preston, R., Shaw, S., 1988. The Warrigundi Igneous Complex: volcanic centres for the Werrie Basalt NSW. New England Orogen: Tectonics and Metallogenesis, pp. 166–171.
Gialanella, P., Heller, F., Haag, M., Nurgaliev, D., Borisov, A., Burov, B., Jasonov, P., Khasanov, D., Ibragimov, S., Zharkov, I., 1997. Late Permian magnetostratigraphy on the eastern Russian platform. Geol. Mijnb. 76 (1-2), 145–154.
Glen, R., 2013. Refining accretionary orogen models for the Tasmanides of eastern Australia. Aust. J. Earth Sci. 60 (3), 315–370.
Glenister, B.F., Boyd, D., Furnish, W., Grant, R., Harris, M., Kozur, H., Lambert, L., Nassichuk, W., Newell, N., Pray, L., 1992. The Guadalupian: proposed international standard for a Middle Permian series. Int. Geol. Rev. 34 (9), 857–888.
Gose, W., Helsley, C., 1972. Paleomagnetic and rock-magnetic studies of the Permian cutler and elephant canyon formations in Utah. J. Geophys. Res. 77 (8), 1534–1548.
Granot, R., Dyment, J., Gallet, Y., 2012. Geomagnetic field variability during the Cretaceous normal superchron. Nat. Geosci. 5 (3), 220–223.
Grybowski, D., 1992. Exploration in permit NSW/P10 in the offshore Sydney Basin. APEA J. 32, 251-251.
Halverson, E., Lowandowski, M., Jelenska, M., 1989. Paleomagnetism of the upper Caroniferous. Phys. Earth Planet. Inter. 55, 54–64.
Harper, L.F., 1915. Geology and Mineral Resources of the Southern Coalfield; with Maps and Sections. WA Gullick, government printer.
Harrington, H., Korsch, R., 1985. Tectonic model for the Devonian to middle Permian of the New England orogen. Aust. J. Earth Sci. 32 (2), 163–179.
Helsley, C., 1965. Paleomagnetic results from the Lower Permian Dunkard series of West Virginia. J. Geophys. Res. 70 (2), 413–424.
Holcombe, R., Stephens, C., Fielding, C., Gust, D., Little, T., Sliwa, R., Kassan, J., McPhie, J., Ewart, A., 1997. Tectonic evolution of the Northern New England Fold Belt: the Permian–Triassic Hunter–Bowen event. Tectonics and Metallogenesis of the New England Orogen 19, pp. 52–65.
Hounslow, M.W., Balabanov, Y.P., 2016. A Geomagnetic Polarity Timescale for the Permian, Calibrated to Stage Boundaries. Geological Society of London.
Hounslow, M.W., McIntosh, G., Edwards, R.A., Laming, D.J., Karloukovski, V., 2016. End of the Kiaman superchron in the Permian of SW England: magnetostratigraphy of the Aylesbeare mudstone and Exeter groups. J. Geol. Soc. 2015–2141.
Irving, E., 1963. Paleomagnetism of the Narrabeen chocolate shales and the Tasmanian dolerite. J. Geophys. Res. 68 (8), 2283–2287.
Irving, E., 1977. Drift of the major continental blocks since the Devonian. Nature 270, 304–309.
Irving, E., Monger, J., 1987. Preliminary paleomagnetic results from the Permian Asitka group, British Columbia. Can. J. Earth Sci. 24 (7), 1490–1497.
Irving, E., Parry, L., 1963. The magnetism of some Permian rocks from new South Wales. Geophys. J. Int. 7 (4), 395–411.
Jelínek, V., 1981. Characterization of the magnetic fabric of rocks. Tectonophysics 79 (3-4), T63–T67.
Jelínek, V., Kropáček, R.V., 1978. Statistical processing of anisotropy of magnetic susceptibility measured on groups of specimens. Stud. Geophys. Geod. 22 (1), 50–62.
Jones, C.H., 2002. User-driven integrated software lives: "Paleomag" paleomagnetics analysis on the Macintosh. Comput. Geosci. 28 (10), 1145–1151.
Jourdan, F., Renne, P.R., 2007. Age calibration of the fish canyon sanidine 40 Ar/39 Ar dating standard using primary K–Ar standards. Geochim. Cosmochim. Acta 71 (2), 387–402.
Kasuya, A., Isozaki, Y., Igo, H., 2012. Constraining paleo-latitude of a biogeographic boundary in mid-Panthalassa: Fusuline province shift on the Late Guadalupian (Permian) migrating seamount. Gondwana Res. 21 (2–3), 611–623.
Khramov, A.N., 1982. Paleomagnetic directions and pole positions: Data for the USSR. Soviet Geophysical Committee. World Data Center-B, Moscow Issue 5, Catalogue.
Khramov, A., Slautsitais, I., 1982. Paleomagnetic directions and pole positions: data for the USSR. Soviet Geophysical Committee, World Data Center-B (Moscow), p. 5.
Khramov, A.N., Tarling, D.H., 1987. Paleomagnetology. Springer Verlag.
Kirschvink, J., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J. Int. 62 (3), 699–718.
Kirschvink, J.L., Isozaki, Y., Shibuya, H., Otofuji, Y.-I., Raub, T.D., Hilburn, I.A., Kasuya, T., Yokoyama, M., Bonifacie, M., 2015. Challenging the sensitivity limits of paleomagnetism: magnetostratigraphy of weakly magnetized Guadalupian–Lopingian (Permian) limestone from Kyushu, Japan. Palaeogeogr. Palaeoclimatol. Palaeoecol. 418, 75–89.
Klootwijk, C., 1985. Palaeomagnetism of the Tasman Fold Belt: indication for mid-Carboniferous large-scale southward displacement of the New England region. Geological Society of Australia Abstracts, pp. 124–127.
Klootwijk, C., 2002. Carboniferous palaeomagnetism of the Rocky Creek block, northern Tamworth Belt, and the New England pole path. Aust. J. Earth Sci. 49 (2), 375–405.
Klootwijk, C., 2003. Carboniferous palaeomagnetism of the Werrie block, northwestern Tamworth Belt, and the New England pole path. Aust. J. Earth Sci. 50 (6), 865–902.
Klootwijk, C.T., Bingham, D.K., 1980. The extent of greater India, III. Palaeomagnetic data from the Tibetan sedimentary series, Thakkhola region, Nepal Himalaya. Earth Planet. Sci. Lett. 51 (2), 381–405.
Kodama, K., 1997. A successful rock magnetic technique for correctng paleomagnetic inclination shallowing: case study of the Nacimiento formation, New Mexico. J. Geophys. Res. Solid Earth 102 (B3), 5193–5205.
Koppers, A.A., 2002. ArArCALC—software for 40 Ar/39 Ar age calculations. Comput. Geosci. 28 (5), 605–619.
Korsch, R., Totterdell, J., Cathro, D., Nicoll, M., 2009a. Early Permian east Australian rift system. Aust. J. Earth Sci. 56 (3), 381–400.
Korsch, R., Totterdell, J., Fomin, T., Nicoll, M., 2009b. Contractional structures and deformational events in the Bowen, Gunnedah and Surat Basins, eastern Australia. Aust. J. Earth Sci. 56 (3), 477–499.
Lackie, M., Schmidt, P., 1993. Remagnetisation of strata during the Hunter-Bowen orogeny. Explor. Geophys. 24 (2), 269–274.
Lackie, M., Schmidt, P., 1996. Thermal events in the Sydney-Bowen basin as defined by palaeomagnetism. Geological Society of Australia Abstracts. Geological Society of Australia, pp. 319–328.
Lanci, L., Tohver, E., Wilson, A., Flint, S., 2013. Upper Permian magnetic stratigraphy of the lower Beaufort group, Karoo Basin. Earth Planet. Sci. Lett. 375, 123–134 (0).
Landenberger, B., Farrell, T., Offler, R., Collins, W., Whitford, D., 1995. Tectonic implications of Rb-Sr biotite ages for the Hillgrove plutonic suite, New England Fold Belt, NSW, Australia. Precambrian Res. 71 (1), 251–263.
Lee, J.-Y., Marti, K., Severinghaus, J.P., Kawamura, K., Yoo, H.-S., Lee, J.B., Kim, J.S., 2006. A redetermination of the isotopic abundances of atmospheric Ar. Geochim. Cosmochim. Acta 70 (17), 4507–4512.
Leven, E.J., 2001. On possibility of using the global Permian stage scale in the Tethyan region. Stratigr. Geol. Correl. 9 (2), 118–131.
Lhuillier, F., Gilder, S.A., Wack, M., He, K., Petersen, N., Singer, B.S., Jicha, B.R., Schaen, A.J., Colon, D., 2016. More stable yet bimodal geodynamo during the Cretaceous superchron? Geophys. Res. Lett.
Li, P., Rosenbaum, G., 2014. Does the manning orocline exist? New structural evidence from the inner hinge of the manning orocline (eastern Australia). Gondwana Res. 25 (4), 1599–1613.
Li, P., Rosenbaum, G., Rubatto, D., 2012. Triassic asymmetric subduction rollback in the southern New England orogen (eastern Australia): the end of the Hunter-Bowen orogeny. Aust. J. Earth Sci. 59 (6), 965–981.
Li, P., Rosenbaum, G., Vasconcelos, P., 2014. Chronological constraints on the Permian geodynamic evolution of eastern Australia. Tectonophysics 617, 20–30.
Llambías, E.J., Quenardelle, S., Montenegro, T., 2003. The Choiyoi group from Central Argentina: a subalkaline transitional to alkaline association in the craton adjacent to the active margin of the Gondwana continent. J. S. Am. Earth Sci. 16 (4), 243–257.
López-Gamundí, O.R., Espejo, I.S., Conaghan, P.J., Powell, C.M., Veevers, J., 1994. Southern South America. Geol. Soc. Am. Mem. 184, 281–330.
Lowrie, W., Alvarez, W., Silva, I.P., Monechi, S., 1980. Lower Cretaceous magnetic stratigraphy in Umbrian pelagic carbonate rocks. Geophys. J. Int. 60 (2), 263–281.
Lozovsky, V.R., Molostovsky, E., 1993. Constructing the early Triassic magnetic polarity time scale. The non marine Triassic. N. M. Mus. Nat. Hist. Sci. Bull. 3, 297–300.
McElhinny, M.W., McFadden, P.L., 2000. Paleomagnetism: Continents and Oceans. Academic Press.
McElhinny, M.W., Powell, C.M., Pisarevsky, S.A., 2003. Paleozoic terranes of eastern Australia and the drift history of Gondwana. Tectonophysics 362 (1), 41–65.
McFadden, P., McElhinny, M., 1988. The combined analysis of remagnetization circles and direct observations in palaeomagnetism. Earth Planet. Sci. Lett. 87 (1), 161–172.
McFadden, P., McElhinny, M., 1990. Classification of the reversal test in palaeomagnetism. Geophys. J. Int. 103 (3), 725–729.
McFadden, P., Ma, X., McElhinny, M., Zhang, Z.K., 1988. Permo-Triassic magnetostratigraphy in China: northern Tarim. Earth Planet. Sci. Lett. 87 (1-2), 152–160.
McQueen, K., 2002. Constraining the weathering history of the Cobar Region, Western NSW, Geological Society of Australia Abstracts. 1999. Geological Society of Australia, p. 426-426.
Menning, M., 2000. Magnetostratigraphic results from the Middle Permian type section, Guadalupe Mountains, West Texas. Permophiles 37, 16.
Menning, M., Katzung, G., Lutzner, H., 1988. Magnetostratigraphic investigations in the Rotliegendes (300–252 MA) of Central-Europe. Z. Geol. Wiss. 16 (11-12) (1045-&).
Menning, M., Alekseev, A., Chuvashov, B., Davydov, V., Devuyst, F.-X., Forke, H., Grunt, T., Hance, L., Heckel, P., Izokh, N., 2006. Global time scale and regional stratigraphic reference scales of central and West Europe, East Europe, Tethys, South China, and North America as used in the Devonian–Carboniferous–Permian Correlation Chart 2003 (DCP 2003). Palaeogeogr. Palaeoclimatol. Palaeoecol. 240 (1), 318–372.
Metcalfe, I., Crowley, J., Nicoll, R., Schmitz, M., 2015. High-precision U-Pb CA-TIMS calibration of middle Permian to lower Triassic sequences, mass extinction and extreme climate-change in eastern Australian Gondwana. Gondwana Res. 28 (1), 61–81.
Miller, J.D., Opdyke, N.D., 1985. Magnetostratigraphy of the Red Sandstone Creek section-Vail, Colorado. Geophys. Res. Lett. 12 (3), 133–136.
Moffitt, R., 2000. A compilation of the geology of the Southern Coalfield; Notes to accompany the 1: 100 000 Southern Coalfield Geology Map. Geological Survey Report No. GS1998, 277.
Morel, P., Irving, E., 1981. Paleomagnetism and the evolution of Pangea. J. Geophys. Res. Solid Earth 86 (B3), 1858–1872.
Musgrave, R.J., 2015. Oroclines in the Tasmanides. J. Struct. Geol. 80, 72–98.
Muttoni, G., Kent, D.V., Garzanti, E., Brack, P., Abrahamsen, N., Gaetani, M., 2003. Early Permian Pangea 'B' to late Permian Pangea 'a'. Earth Planet. Sci. Lett. 215 (3–4), 379–394.

Muttoni, G., Gaetani, M., Kent, D.V., Sciunnach, D., Angiolini, L., Berra, F., Garzanti, E., Mattei, M., Zanchi, A., 2009. Opening of the neo-Tethys Ocean and the Pangea B to Pangea a transformation during the Permian. GeoArabia 14 (4), 17–48.
Nawrocki, J., 1997. Permian to Early Triassic magnetostratigraphy from the central European Basin in Poland: implications on regional and worldwide correlations. Earth Planet. Sci. Lett. 152 (1), 37–58.
Nott, J., Young, R., Idnurm, M., 1991. Sedimentology, weathering, age and geomorphological significance of tertiary sediments on the far south coast of new South Wales. Aust. J. Earth Sci. 38 (3), 357–373.
Ogg, J.G., Ogg, G., Gradstein, F.M., 2016. A Concise Geologic Time Scale: 2016. Elsevier.
Opdyke, M.D., Channell, J.E., 1996. Magnetic Stratigraphy. Academic Press.
Opdyke, N., Roberts, J., Claoué-Long, J., Irving, E., Jones, P., 2000. Base of the Kiaman: its definition and global stratigraphic significance. Geol. Soc. Am. Bull. 112 (9), 1315–1341.
Opdyke, N., Mushayandebvu, M., De Wit, M., 2001. A new palaeomagnetic pole for the Dwyka system and correlative sediments in sub-Saharan Africa. J. Afr. Earth Sci. 33 (1), 143–153.
Peterson, D., Nairn, A., 1971. Palaeomagnetism of Permian redbeds from the South-Western United States. Geophys. J. Int. 23 (2), 191–205.
Phillips, D., Matchan, E., 2013. Ultra-high precision 40 Ar/39 Ar ages for fish canyon tuff and Alder Creek rhyolite sanidine: new dating standards required? Geochim. Cosmochim. Acta 121, 229–239.
Pillans, B., 2002. Climate-driven Weathering Episodes During the Last 200 Million Years in Southern Australia, Geological Society of Australia Abstracts. Geological Society of Australia 1999 p. 428-428.
Pisarevsky, S., 2005. New edition of the global paleomagnetic database. Eos Trans. AGU 86 (17), 170-170.
Pisarevsky, S.A., Rosenbaum, G., Shaanan, U., Hoy, D., Speranza, F., Mochales, T., 2016. Paleomagnetic and geochronological study of Carboniferous forearc basin rocks in the Southern New England orogen (Eastern Australia). Tectonophysics 681, 263–277.
Rapalini, A.E., Fazzito, S., Orué, D., 2006. A new Late Permian paleomagnetic pole for stable South America: the Independencia group, eastern Paraguay. Earth Planets Space 58 (10), 1247–1253.
Renne, P.R., Balco, G., Ludwig, K.R., Mundil, R., Min, K., 2011. Response to the comment by WH Schwarz et al. on "joint determination of 40 K decay constants and 40 Ar*/40 K for the fish canyon sanidine standard, and improved accuracy for 40 Ar/39 Ar geochronology" by PR Renne et al.(2010). Geochim. Cosmochim. Acta 75 (17), 5097–5100.
Renner, J.M., 1988. Placement and paleogeographic significance of Permian-Triassic boundary, Central and Southeast Wyoming. Masters Abstracts International.
Ringis, J., Hawkins, L., Seedsman, K., 1970. Offshore seismic and magnetic surveys of the southern coalfields off Stanwell Park. Australas. Inst. Min. Metall. Proc. 234, 7–16.
Roberts, J., Claoué-Long, J., Foster, C., 1996. SHRIMP zircon dating of the Permian system of eastern Australia*. Aust. J. Earth Sci. 43 (4), 401–421.
Roberts, J., Offler, R., Fanning, M., 2006. Carboniferous to Lower Permian stratigraphy of the southern Tamworth Belt, southern New England orogen, Australia: boundary sequences of the Werrie and Rouchel blocks. Aust. J. Earth Sci. 53 (2), 249–284.
Rogers, J.J., Santosh, M., 2003. Supercontinents in earth history. Gondwana Res. 6 (3), 357–368.
Rosenbaum, G., Li, P., Rubatto, D., 2012. The contorted New England orogen (eastern Australia): new evidence from U-Pb geochronology of early Permian granitoids. Tectonics 31 (1).
Sakai, H., Kikawa, E., Offle, R., Zwingmann, H., 2005. Paleomagnetic and K-Ar dating studies of Late Cretaceous basaltic dykes intruding Late Permian sediments of Newcastle coal measures, Newcastle, Australia-relationship to the opening of the Tasman Sea. Polar Geosci. 18, 58–66.
Schaltegger, U., Brack, P., 2007. Crustal-scale magmatic systems during intracontinental strike-slip tectonics: U, Pb and Hf isotopic constraints from Permian magmatic rocks of the southern alps. Int. J. Earth Sci. 96 (6), 1131–1151.
Scheibner, E., 1974. Fossil fracture zones (transform faults), segmentation and correlation problems in the Tasman Fold Belt System. The Tasman Geosyncline–A Symposium in Honour of Professor Dorothy Hill, Queensland Division Geological Society of Australia, Brisbane, pp. 65–96.
Schmidt, P., Embleton, B., 1981. Magnetic overprinting in southeastern Australia and the thermal history of its rifted margin. J. Geophys. Res. Solid Earth 86 (B5), 3998–4008.
Schmidt, P., Williams, G., 2002. Palaeomagnetic Dating of the Hamersley Surface and Deep Weathering in the Pilbara-Northern Yilgarn Region, Western Australia, Geological Society of Australia Abstracts. Geological Society of Australia 1999 p. 431-431.
Shaanan, U., Rosenbaum, G., Li, P., Vasconcelos, P., 2014. Structural evolution of the early Permian Nambucca block (New England orogen, eastern Australia) and implications for oroclinal bending. Tectonics 33 (7), 1425–1443.
Shaanan, U., Rosenbaum, G., Pisarevsky, S., Speranza, F., 2015a. Paleomagnetic data from the New England orogen (eastern Australia) and implications for oroclinal bending. Tectonophysics 664, 182–190.
Shaanan, U., Rosenbaum, G., Wormald, R., 2015b. Provenance of the Early Permian Nambucca block (eastern Australia) and implications for the role of trench retreat in accretionary orogens. Geol. Soc. Am. Bull. 127 (7-8), 1052–1063.
Sinito, A.M., Valencio, D.A., Vilas, J.F., 1979. Palaeomagnetism of a sequence of Upper Palaeozoic—Lower Mesozoic red beds from Argentina. Geophys. J. Int. 58 (2), 237–247.
Smith, A.G., Hallam, A., 1970. The Fit of the Southern Continents.
Somoza, R., Zaffarana, C.B., 2008. Mid-Cretaceous polar standstill of South America, motion of the Atlantic hotspots and the birth of the Andean cordillera. Earth Planet. Sci. Lett. 271 (1), 267–277.
Steiner, M.B., 2006. The magnetic polarity time scale across the Permian-Triassic boundary. Geol. Soc. Lond., Spec. Publ. 265 (1), 15–38.
Steiner, M.B., Morales, M., Shoemaker, E.M., 1993. Magnetostratigraphic, biostratigraphic, and lithologic correlations in Triassic strata of the western United States. Special Publication-Society of Economic Paleontologists and Mineralogists 49, p. 41-41.
Sun, W.W., Kodama, K., 1992. Magnetic anisotropy, scanning electron microscopy, and X ray pole figure goniometry study of inclination shallowing in a compacting clay-rich sediment. J. Geophys. Res. Solid Earth 97 (B13), 19599–19615.
Symons, D., 1990. Early Permian pole: evidence from the Pictou red beds, Prince Edward Island, Canada. Geology 18 (3), 234–237.
Szurlies, M., Bachmann, G.H., Menning, M., Nowaczyk, N.R., Käding, K.-C., 2003. Magnetostratigraphy and high-resolution lithostratigraphy of the Permian–Triassic boundary interval in Central Germany. Earth Planet. Sci. Lett. 212 (3), 263–278.
Tan, X., Kodama, K., 2002. Magnetic anisotropy and paleomagnetic inclination shallowing in red beds: evidence from the Mississippian Mauch Chunk Formation, Pennsylvania. J. Geophys. Res. Solid Earth 107 (B11).
Tan, X., Kodama, K.P., Fang, D., 2002. Laboratory depositional and compaction-caused inclination errors carried by haematite and their implications in identifying inclination error of natural remanence in red beds. Geophys. J. Int. 151 (2), 475–486.
Tan, X., Kodama, K.P., Gilder, S., Courtillot, V., 2007. Rock magnetic evidence for inclination shallowing in the Passaic Formation red beds from the Newark basin and a systematic bias of the Late Triassic apparent polar wander path for North America. Earth Planet. Sci. Lett. 254 (3–4), 345–357.
Tarduno, J.A., 1990. Brief reversed polarity interval during the Cretaceous normal polarity superchron. Geology 18 (8), 683–686.
Tauxe, L., 2005. Inclination flattening and the geocentric axial dipole hypothesis. Earth Planet. Sci. Lett. 233 (3), 247–261.
Tauxe, L., Kent, D.V., 2004. A simplified statistical model for the geomagnetic field and the detection of shallow bias in paleomagnetic inclinations: was the ancient magnetic field dipolar? Timescales of the Paleomagnetic Field, pp. 101–115
Tauxe, L., Kodama, K.P., Kent, D.V., 2008. Testing corrections for paleomagnetic inclination error in sedimentary rocks: a comparative approach. Phys. Earth Planet. Inter. 169 (1–4), 152–165.
Tohver, E., Lanci, L., Wilson, A., Hansma, J., Flint, S., 2015. Magnetostratigraphic constraints on the age of the lower Beaufort group, western Karoo basin, South Africa, and a critical analysis of existing U-Pb geochronological data. Geochem. Geophys. Geosyst. 16 (10), 3649–3665.
Torsvik, T.H., Van der Voo, R., 2002. Refining Gondwana and Pangea palaeogeography: estimates of Phanerozoic non-dipole (octupole) fields. Geophys. J. Int. 151 (3), 771–794.
Torsvik, T.H., Müller, R.D., Van der Voo, R., Steinberger, B., Gaina, C., 2008. Global plate motion frames: toward a unified model. Rev. Geophys. 46 (3).
Torsvik, T.H., Van der Voo, R., Preeden, U., Mac Niocaill, C., Steinberger, B., Doubrovine, P.V., van Hinsbergen, D.J., Domeier, M., Gaina, C., Tohver, E., 2012. Phanerozoic polar wander, palaeogeography and dynamics. Earth Sci. Rev. 114 (3), 325–368.
Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184 (1), 1–9.
Van der Voo, R., 1993. Paleomagnetism of the Atlantic. Cambridge University Press, Tethys and Iapetus oceans.
Van der Voo, R., French, R., 1974. Apparent polar wandering for the Atlantic-bordering continents: Late Carboniferous to Eocene. Earth Sci. Rev. 10 (2), 99–119.
Van der Voo, R., Torsvik, T.H., 2001. Evidence for late Paleozoic and Mesozoic non-dipole fields provides an explanation for the Pangea reconstruction problems. Earth Planet. Sci. Lett. 187 (1), 71–81.
Van der Voo, R., Peinado, J., Scotese, C.R., 1984. A paleomagnetic reevaluation of Pangea reconstructions, plate reconstruction from Paleozoic paleomagnetism. Am. Geophys. Union 11–26.
VandenBerg, J., Klootwijk, C., Wonders, A., 1978. Late Mesozoic and Cenozoic movements of the Italian peninsula: further paleomagnetic data from the Umbrian sequence. Geol. Soc. Am. Bull. 89 (1), 133–150.
Veevers, J.J., 2006. Updated Gondwana (Permian–Cretaceous) earth history of Australia. Gondwana Res. 9 (3), 231–260.
Veevers, J.J., Powell, C.M., 1994. Permian-Triassic Pangean Basins and Foldbelts along the Panthalassan Margin of Gondwanaland. Geological Society of America.
Verati, C., Jourdan, F., 2014. Modelling effect of sericitization of plagioclase on the 40 K/40Ar and 40Ar/39Ar chronometers: implication for dating basaltic rocks and mineral deposits. Geol. Soc. Lond., Spec. Publ. 378 (1), 155–174.
Watson, G., 1956. A Test for Randomness of Directions. Geophysical Supplements to the Monthly Notices of the Royal Astronomical Society 7(4), pp. 160–161.
Wilde, G.L., 1999. Formal Designation: Reef Trail Member, Bell Canyon Formation, and its Significance for Recognition of the Guadalupian-Lopingian Boundary.
Wynne, P., Irving, E., Osadetz, K., 1983. Paleomagnetism of the Esayoo Formation (Permian) of northern Ellesmere Island: possible clue to the solution of the Nares Strait dilemma. Tectonophysics 100 (1-3), 241–256.